Financial Networks, Credit Reallocation, and Monetary Transmission

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Abstract

Monetary policy transmission depends not only on how much bank credit expands, but on where in the financial network that expansion occurs. We embed interconnected banks into a New Keynesian model and show that policy-induced credit reallocation changes the output-inflation composition of monetary transmission. The key object is an effective network weight, which measures how much each bank's lending moves aggregate financial capacity after valuation feedback propagates through the network. When easing shifts credit toward banks with high effective weights, the same aggregate credit expansion generates larger real effects for a given inflation response. The mechanism also implies stronger network amplification for quantitative easing than for conventional rate policy, because QE loads more directly on bank balance sheets. Finally, past accommodation leaves inherited financial wedges that shape future transmission. The results show that financial networks matter for routine monetary policy, not only for crisis propagation.

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1 Introduction

Empirical work on financial connectedness points to a common lesson: banks' macro-financial importance is not fully summarized by their size or own risk, but also reflects their position in the financial network (Adrian and Brunnermeier 2016; Diebold and Yılmaz 2014). This raises a natural monetary-policy question: whether the aggregate effect of credit expansion depends on which banks supply the marginal credit. In the workhorse New Keynesian model, the demand–supply composition of monetary policy transmission is pinned down by price rigidity and household preferences. The financial sector is either absent or represented by a single intermediary, so the identity of the bank supplying marginal credit has no aggregate content. Monetary policy changes the amount of credit, but the model has no role for the macroeconomic meaning of who supplies it.

In modern banking systems, this abstraction misses a basic feature of financial intermediation. Banks are connected through balance sheets, ownership claims, common exposures, and other valuation linkages. As a result, a dollar of lending by one bank can have a different effect on aggregate financial capacity than a dollar of lending by another bank. Since network positions shape banks' responses to policy shocks, monetary policy changes not only total credit, but also the cross-sectional allocation of credit across institutions. This paper embeds a financial network of n interconnected banks into an otherwise standard New Keynesian DSGE and studies how this financial reallocation channel changes the output–inflation trade-off.

The channel operates through two network margins: heterogeneous bank responses and aggregate network importance. The first margin determines which banks respond most strongly to monetary policy. We summarize this response margin by the dependency index, which is determined by a bank's network position: banks whose own lending feeds back more strongly into their own market value experience larger relaxations of their leverage constraints and therefore expand credit more aggressively after easing. The second margin determines whether those bank-level responses matter for aggregate outcomes. We summarize this aggregation margin by the effective network weight: a bank has a high effective network weight when changes in its fundamental value propagate through the network and move the financial capacity of the banking system as a whole. Monetary transmission is strongest when these two margins are aligned. If easing shifts credit toward banks that are both highly responsive and highly effective in the network, the same policy shock produces a large aggregate financial-capacity response. If the banks that respond most have low effective network weights, policy can substantially reallocate credit across banks while generating only a modest aggregate effect.

Framework. We build a New Keynesian DSGE model with four blocks. The financial sector consists of n banks that choose long-bond positions subject to a value-based

leverage constraint. Bank market values are linked through a dependency matrix $\mathbf{A}$, micro-founded by an equity cross-holdings structure in the spirit of Elliott et al. (2014) and augmented with a friction parameter capturing tax and regulatory costs along inter-bank dividend chains. The household sector follows Sims et al. (2023): a patient saver and an impatient borrower, so credit conditions redistribute consumption across household types. The production sector is the textbook New Keynesian block, with a final-good aggregator over a continuum of intermediate goods produced by monopolistically competitive firms that set prices subject to Calvo frictions. The central bank operates both a Taylor rule and a quantitative easing rule that purchases long-term bonds.

Log-linearizing the economy yields a four-equation system in output, inflation, aggregate financial capacity Θ_t^{net}, and the cross-sectional credit distribution $\hat{\mathbf{b}}_t^{FI}$. The key macroeconomic asymmetry is that aggregate financial capacity enters the IS curve in changes but enters the Phillips Curve in levels. A temporary increase in financial capacity raises borrower demand today relative to tomorrow. At the same time, a higher level of borrower consumption lowers saver consumption for a given level of output, raises the saver's marginal utility of income, and lowers marginal cost through labor supply. The network governs the size of this demand–supply composition effect by determining how bank-level credit movements aggregate into Θ_t^{net}.

Main results. The first result is that the financial reallocation channel changes the output–inflation composition of monetary transmission. Easing reallocates credit toward banks with stronger value feedback, but this reallocation matters for inflation only to the extent that it changes network-weighted financial capacity. The size of the supply-side credit wedge therefore depends on an alignment condition: how closely the banks that respond most to policy line up with the banks that carry large effective network weights. Network connectedness alone is not enough; what matters is whether policy-induced reallocation moves credit toward banks whose lending has large aggregate consequences.

The second result is asymmetric amplification across policy instruments. Network amplification operates on the part of policy transmission that passes through bank balance sheets, spreads, and bond portfolios. Conventional rate policy has such a component, but it also has a direct household Euler-equation channel. QE is more concentrated in the balance-sheet component because it works through bond absorption, bond prices, and inherited bank portfolios. Consequently, since the common balance-sheet component is the network-amplified part and the rate instrument retains a positive household bypass, the output amplification ratio for QE exceeds that for interest rate policy. The result explains how QE can generate sizable real effects with comparatively muted inflation effects without requiring a uniformly stronger unconventional policy instrument.

The third result is financial memory. Monetary policy changes the cross-sectional allocation of bank credit, and this reallocation persists because banks enter subsequent

periods with legacy bond portfolios. The inherited cross-section of bond holdings affects network financial capacity: the same aggregate stock of bonds can support different financial capacity depending on which banks hold it. Past accommodation therefore changes the financial position from which future policy shocks are transmitted. In the first-order system, this memory operates through inherited financial wedges rather than through time-varying local coefficients; richer state-dependent versions would allow market shares or effective network weights themselves to evolve.

Related literature. The closest macroeconomic benchmark is Sims et al. (2023), who show how a representative intermediary and borrower-saver heterogeneity generate a four-equation New Keynesian system in which credit affects both the IS curve and the Phillips Curve. We inherit this aggregate structure but replace their representative financial intermediary with a financial network of interconnected banks. This makes the cross-section of credit supply a macroeconomic state variable. More broadly, the paper builds on DSGE models with financial frictions (Brunnermeier and Sannikov 2014; Gertler and Karadi 2011; Gertler and Kiyotaki 2010, 2015), including work on monetary reversals (Abadi et al. 2023), central-bank toolkits (Sims and Wu 2021), and conventional versus unconventional policy operations (Bianchi and Bigio 2022; Wu and Xie 2025). The distinction is that these models generally aggregate the banking sector, whereas our channel depends on which banks supply marginal credit.

The paper also relates to work on bank heterogeneity and supply-side monetary transmission. Bellifemine et al. (2025) show that heterogeneous bank lending propensities amplify aggregate demand, while Jamilov and Monacelli (2025) generates bank wealth heterogeneity from idiosyncratic histories. Our heterogeneity is structural and network based: balance-sheet linkages determine both the banks that respond to policy and the aggregate importance of their lending. Ottonello and Winberry (2020) show that cross-sectional financial positions are state variables for monetary transmission on the borrower side. We obtain an analogous state variable on the intermediary side. On the supply side, Baqaee et al. (2024) derive monetary supply effects from heterogeneous firm pass-through and markups, while Gödl-Hanisch (2025) links bank concentration to Phillips Curve flattening. Our supply effect instead comes from credit-induced consumption redistribution between borrowers and savers, with its strength governed by the banking network.

Finally, the paper contributes to the networks-in-macroeconomics literature. Production-network models show how input-output linkages make aggregate fluctuations and the output–inflation trade-off depend on network structure (Acemoglu et al. 2012; La'O and Tahbaz-Salehi 2022; Rubbo 2023). Financial-network models study how cross-holdings and interbank linkages transmit distress and amplify shocks (Acemoglu et al. 2015; Elliott et al. 2014). We bring the financial-network object into routine monetary transmission rather than crisis propagation. The core-periphery motivation is supported by Farboodi

(2023), and Glasserman and Young (2016) and Jackson and Pernoud (2021) provide surveys. Empirically, Adrian and Brunnermeier (2016) document that a bank's own risk and its systemic contribution are only weakly correlated, consistent with the distinction between market shares and effective network weights that activates our mechanism.

2 A New-Keynesian DSGE with a Bank Dependency Network

This section develops a New Keynesian DSGE model populated by three types of agents: a financial sector of n banks connected through a dependency network (Section 2.1), two households—a patient saver and an impatient borrower (Section 2.2), and a central bank conducting both conventional and unconventional monetary policy (Section 2.3). The production sector follows a standard Calvo pricing structure; its details, together with market clearing conditions and the formal equilibrium definition, are collected in Appendix C.

The financial sector is the paper's main innovation. Banks are tied together through an equity cross-holdings network whose reduced-form object—the *dependency matrix* $\mathbf{A}$—maps bank-level fundamental values into market values. The cross-holdings matrix $\mathbf{C}$ serves as our micro-foundation: it is the simplest ownership structure that generates, from first principles, a dependency matrix with the properties required by the subsequent theory—non-negativity, a core-periphery pattern in the diagonal elements, value dissipation under frictions, and a convergent Neumann expansion. Its role is analogous to the Leontief production function in the input-output literature: not a literal description of every real-world linkage, but a parsimonious structure that produces the correct reduced-form object. All analytical results depend only on properties of $\mathbf{A}$, never on the specific form of $\mathbf{C}$.

2.1 The Financial Sector

Environment. Consider a discrete-time economy with an infinite horizon, $t = 0, 1, 2, \ldots$ The financial sector consists of n intermediaries (banks) indexed by $i \in \mathcal{N} = \{1, \ldots, n\}$. Banks are risk-neutral and choose portfolios to maximize their market value.

Financial assets. Banks invest in two asset classes. The first is a long-term risky bond modeled as a perpetuity with geometrically decaying coupons: a bond issued at time t pays 1 unit at $t+1$, κ units at $t+2$, κ^2 at $t+3$, and so on, where $\kappa \in [0, 1]$ is the decay parameter. Let Q_t denote the nominal bond price. The gross holding-period

return realized at $t+1$ is

$$R^b_{t+1} = \frac{1+\kappa Q_{t+1}}{Q_t}. \tag{1}$$

The second asset is risk-free reserves $RE_{i,t}$, earning a gross nominal rate R^{re}_t between t and $t+1$. The first order condition for reserve holding implies $\mathbb{E}_t[M^N_{t,t+1}(R^{re}_t - R^s_t)] = 0$, where $M^N_{t,t+1}$ is the nominal stochastic discount factor and R^s_t is the deposit rate. Reserves therefore yield no excess return in expectation, and the bank's problem reduces to managing the risky bond portfolio.

Banker demographics. Following Sims et al. (2023) and Wu and Xie (2025), we assume that each banker is born at the beginning of period t, operates for one period, and exits at the beginning of $t+1$ with probability one. Upon exit, the banker distributes terminal wealth as dividends. The newly born FIs receive an exogenous transfer from the unconstrained household including two components: (i) a fixed amount of new equity and (ii) the outstanding long-term bonds held by the previous intermediary. [1]

2.1.1 The Financial Network

The central object in our model is the *dependency matrix* $\mathbf{A}$, which maps bank-level fundamental values to market values and thereby governs how the financial network shapes lending incentives. This subsection derives $\mathbf{A}$ directly in the presence of the structural friction that is central to our analysis.

Cross-holdings and ownership structure. The network is captured by a matrix $\mathbf{C} \in \mathbb{R}^{n\times n}_+$, where $C_{ij} \in [0,1)$ denotes the fraction of bank j's equity held by bank i. We impose three structural restrictions. First, *no self-ownership*: $C_{ii} = 0$ for all i. Second, *positive outside ownership*: the residual share of bank i held by outside investors (households) is $\hat{C}_{ii} \equiv 1 - \sum_{j\in\mathcal{N}} C_{ji} \in (0,1]$ for all i. Third, *convergence*: the spectral radius $\rho(\mathbf{C}) < 1$, guaranteeing that the matrix $(\mathbf{I}-\mathbf{C})^{-1}$ exists and has a convergent Neumann series expansion. Let $\hat{\mathbf{C}} \equiv \mathrm{diag}(\hat{C}_{11},\ldots,\hat{C}_{nn})$ collect the outside ownership shares. The matrix $\hat{\mathbf{C}}$ records the physical fact of who holds equity in whom and does not depend on any friction parameter.

The friction parameter τ. We introduce a structural friction parameter $\tau \in [0,1)$ associated with interbank equity linkages. When dividends flow through cross-holding chains, each link transmits only a fraction $(1-\tau)$ of the dividend it receives. This friction captures two sources of value attenuation in interbank ownership chains. First, *tax leakage*: under U.S. tax law (26 U.S.C. §243), the Dividends Received Deduction for portfolio

[1] While individual bankers turn over each period, the institutional structure of each bank—its position in the financial network and its inherited bond portfolio—persists across generations.

holdings below 20% is limited to 50%, yielding an effective tax wedge of approximately 10.5% at the current federal corporate rate of 21%. Second, *regulatory capital costs*: Basel III mandates that significant investments in other financial institutions' capital be deducted from CET1 or assigned punitive risk weights, imposing an opportunity cost on cross-holding positions. The conceptual distinction between $\hat{C}_{ii}$ and τ is sharp: the former records the ownership structure—the physical fact of who holds equity in whom—while the latter governs the efficiency of dividend transmission through the network.

Fundamental value. Each bank i maintains a balance sheet consisting of bonds $B_{i,t}$, reserves $RE_{i,t}$, deposits $S_{i,t}$, and an exogenous equity endowment $E_{i,t}$:

$$Q_t B_{i,t} + RE_{i,t} = S_{i,t} + E_{i,t}. \tag{2}$$

The bank's net cash flow (dividend) upon exit at $t+1$ is:

$$D_{i,t+1} = \underbrace{(R^b_{t+1} - R^s_t)\, Q_t B_{i,t}}_{\text{credit spread income}} + \underbrace{(R^{re}_t - R^s_t)\, RE_{i,t}}_{=0 \text{ in expectation}} + \underbrace{R^s_t\, E_{i,t}}_{\text{endowment return}}. \tag{3}$$

The fundamental value of bank i is the expected discounted dividend from its own portfolio operations, absent any network effects:

$$x_{i,t} \equiv \mathbb{E}_t\big[M^N_{t,t+1}\, D_{i,t+1}\big] = \Delta_t\, Q_t B_{i,t} + \bar{x}_{i,t}, \tag{4}$$

where $\Delta_t \equiv \mathbb{E}_t[M^N_{t,t+1}(R^b_{t+1} - R^s_t)]$ is the aggregate credit spread and $\bar{x}_{i,t} \equiv \mathbb{E}_t[M^N_{t,t+1} R^s_t]\, E_{i,t}$ is the value of the endowment. Let $\mathbf{x}_t \equiv (x_{1,t}, \ldots, x_{n,t})'$ collect fundamentals.

Total equity value with friction. A bank's total equity value includes its fundamental assets and the value of its equity stakes in other banks, after accounting for the friction τ that attenuates dividend transmission through each link:

$$\mathbf{V}_t = \mathbf{x}_t + (1-\tau)\, \mathbf{C}\, \mathbf{V}_t, \tag{5}$$

Solving the fixed-point equation:

$$\mathbf{V}_t = \big[\mathbf{I} - (1-\tau)\mathbf{C}\big]^{-1} \mathbf{x}_t = \left(\sum_{k=0}^{\infty} (1-\tau)^k \mathbf{C}^k \right) \mathbf{x}_t. \tag{6}$$

The Neumann series has a transparent economic interpretation: $(1-\tau)^k \mathbf{C}^k \mathbf{x}_t$ captures the value flowing to each bank through ownership chains of exactly k links, attenuated by a factor $(1-\tau)^k$. A direct holding ($k=1$) transmits value with one round of friction; an indirect holding through an intermediary ($k=2$) transmits value with two rounds of

friction; and so on.

Market value and the dependency matrix. Outside investors hold a fraction $\hat{C}_{ii}$ of each bank's total equity. The market value of bank i's outside equity—the value relevant for the bank's objective and its leverage constraint—is:

$$\mathbf{v}_t = \hat{\mathbf{C}}\,\mathbf{V}_t = \underbrace{\hat{\mathbf{C}}\left[\mathbf{I} - (1-\tau)\mathbf{C}\right]^{-1}}_{\equiv\,\mathbf{A}(\tau)}\,\mathbf{x}_t. \tag{7}$$

Definition 1 (Dependency Matrix)**.** *The* dependency matrix $\mathbf{A}(\tau) \equiv \hat{\mathbf{C}}\,[\mathbf{I} - (1-\tau)\mathbf{C}]^{-1}$ *maps fundamental values to market values. Its element A_{ij} measures the marginal effect of a unit increase in bank j's fundamental value on bank i's market value. The diagonal element A_{ii}, which we term bank i's* dependency index*, captures how much of the bank's own fundamental value is retained in its market value after accounting for all network interactions and frictions.*

For notational economy, we suppress the dependence on τ except where comparative statics are explicitly discussed.

In the log-linear equilibrium system, the level derivative $\mathbf{A}$ is converted into the valuation-elasticity matrix

$$\mathcal{A}_{ij} \equiv \frac{A_{ij}\bar{x}_j}{\bar{v}_i}, \qquad \hat{v}_{i,t} = \sum_j \mathcal{A}_{ij}\hat{x}_{j,t}. \tag{8}$$

Thus $\mathbf{A}$ is the level dependency matrix, while $\mathcal{A}$ is the object that enters the log-linear four-equation system.

Remark 1 (Two Forces Shaping A_{ii})**.** *The dependency index is the product of two opposing forces. Expanding A_{ii} using the Neumann series:*

$$A_{ii} = \underbrace{\hat{C}_{ii}}_{\textit{dilution factor}} \times \underbrace{\left[1 + (1-\tau)^2(C^2)_{ii} + (1-\tau)^3(C^3)_{ii} + \cdots\right]}_{\textit{amplification factor}}, \tag{9}$$

where the $k = 1$ term vanishes because $C_{ii} = 0$, and $(C^k)_{ii} = \sum_{j_1,\ldots,j_{k-1}} C_{ij_1}C_{j_1j_2}\cdots C_{j_{k-1}i}$ sums over all ownership paths of length k that begin and end at bank i. The dilution factor *$\hat{C}_{ii} = 1 - \sum_j C_{ji}$ is smaller for banks that are more heavily held by other banks: when other institutions own large stakes in bank i, a smaller fraction of i's total value accrues to outside investors. The* amplification factor *captures the cumulative intensity of all closed-loop ownership chains returning to bank i, attenuated by $(1-\tau)^k$ at each path length k, and is larger for banks embedded in denser feedback loops. These two forces pull in opposite directions: a core bank (heavily held by many counterparties) has a large amplification factor but a small dilution factor; a peripheral bank has a dilution factor*

close to one but a weak amplification factor. In the frictionless benchmark ($\tau = 0$), the amplification factor operates at full force and can push A_{ii} above one for banks at the center of dense ownership loops. As τ increases, the amplification factor is progressively suppressed, and in the limit $\tau \to 1$, $\mathbf{A} \to \hat{\mathbf{C}}$ so that each bank's dependency index equals its outside ownership share.

The friction τ destroys aggregate value, with the dissipation concentrated at the most interconnected institutions. The following result quantifies this dissipation.

Proposition 1 (Value Dissipation)**.** *For any cross-holdings matrix $\mathbf{C}$ satisfying the structural restrictions above and any $\tau \in [0, 1)$:*

$$\sum_{i \in \mathcal{N}} A_{ij}(\tau) = 1 - \tau \sum_{i \in \mathcal{N}} \left[\mathbf{C}\left(\mathbf{I} - (1-\tau)\mathbf{C}\right)^{-1}\right]_{ij} \leq 1 \quad \textit{for all } j \in \mathcal{N}, \tag{10}$$

with equality if and only if $\tau = 0$ or $\mathbf{C} = \mathbf{0}$.

Proof. See Appendix A.1. □

The economic content is transparent. One dollar of bank j's fundamental value generates less than one dollar of aggregate market value across the banking system whenever $\tau > 0$. The shortfall equals the cumulative tax and regulatory leakage incurred as dividends flow through all direct and indirect ownership chains originating from bank j. In the frictionless case ($\tau = 0$) the column sums equal one exactly: the network redistributes value across banks without dissipating it.

Proposition 1 is not merely a mathematical property of column sums. The heterogeneity in A_{ii} created by $\tau > 0$ is the prerequisite for two central mechanisms in general equilibrium: the credit reallocation channel (Proposition 3) that generates an independent macroeconomic state variable, and the network flattening of the Phillips Curve that constitutes the paper's central contribution.

2.1.2 The Bank's Problem

Each bank i chooses bond holdings $B_{i,t}$ to maximize its market value $v_{i,t}$, subject to a leverage constraint that ties borrowing capacity to the bank's network-determined market value and penalizes portfolio adjustment:

$$Q_t B_{i,t} \leq \phi_{i,t}\, v_{i,t} - \eta_B\, \bar{Q}\, (B_{i,t} - B_{i,t-1}), \tag{11}$$

where $\phi_{i,t} > 0$ is an idiosyncratic credit-capacity shock and $\eta_B \geq 0$ is a portfolio adjustment cost parameter. The adjustment cost term penalizes changes in bond holdings relative to the previous period, reflecting regulatory reporting costs, market impact in illiquid bond markets, and internal risk-management frictions (Gertler and Kiyotaki

2010). The cost is scaled by the steady-state bond price $\bar{Q}$ for dimensional consistency and vanishes in steady state ($B_{i,t} = B_{i,t-1}$), so it does not affect the long-run equilibrium. This specification creates a value feedback loop: lending decisions affect fundamental values through $x_{i,t} = \Delta_t Q_t B_{i,t} + \bar{x}_{i,t}$, which propagate through the network to affect market values $v_{i,t} = \sum_j A_{ij} x_{j,t}$, which in turn determine the tightness of the constraint. The strength of this internal feedback depends on the bank's dependency index A_{ii}.

To obtain a closed-form solution, we impose the following assumption.

Assumption 1 (Fixed Spillovers)**.** *Each bank i internalizes the effect of its lending on its own valuation through the self-loop A_{ii}, but treats the value contributions from other banks as exogenous to its portfolio choice. Formally, bank i decomposes its market value as:*

$$v_{i,t} = A_{ii}\big(\Delta_t Q_t B_{i,t} + \bar{x}_{i,t}\big) + \bar{v}_{i,t}, \tag{12}$$

where $\bar{v}_{i,t} \equiv \sum_{j \neq i} A_{ij} x_{j,t}$ is taken as given.

This assumption is the financial-network analogue of taking competitors' output as given in a Cournot oligopoly: each bank recognizes that its own lending feeds back to its own valuation through the network, but does not attempt to manipulate other banks' valuations strategically. The assumption is exact when banks are small relative to the network and becomes a close approximation when the number of banks is large.

Under Assumption 1, bank i solves:

$$\max_{B_{i,t} \geq 0} \; v_{i,t} = A_{ii}\big(\Delta_t Q_t B_{i,t} + \bar{x}_{i,t}\big) + \bar{v}_{i,t} \quad \text{s.t.} \quad Q_t B_{i,t} \leq \phi_{i,t}\, v_{i,t} - \eta_B\, \bar{Q}\, (B_{i,t} - B_{i,t-1}). \tag{13}$$

Since the objective is linear and increasing in $B_{i,t}$ (given $\Delta_t > 0$), risk-neutral banks lever up until the constraint binds. Substituting the binding constraint and solving yields the optimal credit supply:

$$B^*_{i,t} = \frac{\phi_{i,t}\big(A_{ii}\, \bar{x}_{i,t} + \bar{v}_{i,t}\big) + \eta_B\, \bar{Q}\, B_{i,t-1}}{Q_t\big(1 - \phi_{i,t}\, A_{ii}\, \Delta_t\big) + \eta_B\, \bar{Q}}. \tag{14}$$

The adjustment cost parameter η_B introduces inertia: current lending depends positively on the inherited portfolio $B_{i,t-1}$. In steady state, $B_{i,t} = B_{i,t-1}$ and the adjustment cost term cancels from both sides, so the steady-state allocation is identical to the frictionless case ($\eta_B = 0$). For the analytical results that follow, we present expressions in the $\eta_B = 0$ case for clarity; all results extend to $\eta_B > 0$ with the feedback multiplier replaced by $(1 - \phi_{i,t} A_{ii} \Delta_t + \eta_B \bar{Q}/Q_t)^{-1}$, which dampens contemporaneous amplification but preserves the comparative static $\partial \mathcal{Z}_{i,t}/\partial A_{ii} > 0$.

Assumption 2 (Stability)**.** *The feedback effect is bounded: $1 - \phi_{i,t}\, A_{ii}\, \Delta_t > 0$ for all i and t. For the log-linear system, the matrix feedback is bounded by $\rho((1+\eta_B)^{-1}\mathcal{A}\mathbf{S}) < 1$.*

Assumption 2 ensures that the value feedback loop converges rather than explodes: the amplification from one round of the feedback is less than one. It is the bank-level analogue of requiring that a geometric series has ratio less than one.

Before embedding the banking sector into general equilibrium, it is useful to isolate the micro-level mechanisms through which the network structure shapes credit supply decisions. The following definition formalizes the partial-equilibrium environment.

Definition 2 (Partial Equilibrium)**.** *Given the network structure* $\mathbf{A}$*, exogenous processes* $\{\Delta_t, \phi_{i,t}, \bar{x}_{i,t}\}$*, and asset prices* $\{Q_t\}$*, a partial equilibrium is a collection of portfolio choices* $\{B^*_{i,t}\}_{i\in\mathcal{N}}$ *and valuations* $\{\mathbf{v}_t, \mathbf{x}_t\}$ *such that: (i) each bank* i *chooses* $B^*_{i,t}$ *according to* (14)*; (ii) fundamental values satisfy* $x_{i,t} = \Delta_t\, Q_t B^*_{i,t} + \bar{x}_{i,t}$*; and (iii) market values satisfy* $\mathbf{v}_t = \mathbf{A}\,\mathbf{x}_t$*.*

2.1.3 Credit Supply in Partial Equilibrium

In this subsubsection, we take asset prices $\{Q_t\}$, the nominal stochastic discount factor $M^N_{t,t+1}$, and the aggregate credit spread Δ_t as exogenous, and derive analytical results that characterize how the dependency matrix shapes individual bank responses and aggregate credit market structure. These partial-equilibrium insights—combined with the household and monetary policy blocks that follow—determine the full general equilibrium dynamics analyzed in Sections 3–4.

The value feedback channel. The closed-form solution (14) reveals that the network enters each bank's credit decision through a single channel: the dependency index A_{ii} appears in the denominator as part of a feedback multiplier $\left(1 - \phi_{i,t}\, A_{ii}\, \Delta_t\right)^{-1}$. Banks with higher A_{ii} face a more powerful feedback loop—each dollar of lending generates more market value, which relaxes the leverage constraint by more, enabling more lending—and therefore respond more aggressively to changes in the credit spread.

Proposition 2 (Value Feedback Channel)**.** *Under Assumptions 1–2, the semi-elasticity of bank* i*'s credit supply with respect to the aggregate credit spread* Δ_t *is strictly positive and strictly increasing in the dependency index* A_{ii}*. Banks that retain a larger share of their own fundamental value respond more aggressively to the same policy impulse.*

Proof. See Appendix A.2. The formal expression is $\mathcal{Z}_{i,t} = \phi_{i,t}A_{ii}/(1 - \phi_{i,t}A_{ii}\Delta_t)$, which is manifestly positive and increasing in A_{ii}. □

The economic mechanism operates as follows. An expansionary monetary policy that widens the credit spread Δ_t increases the expected profitability of lending for all banks. For a standalone bank ($A_{ii} = 1$, no network), this raises its fundamental value, which relaxes its leverage constraint by a proportional amount, enabling some additional lending. For a bank embedded in the network, the initial gain in fundamental value propagates

through the dependency network to partner banks, whose higher valuations in turn raise the value of bank i's equity stakes in them, feeding the gain back to bank i. The strength of this self-reinforcing loop depends on A_{ii}: it captures how efficiently the feedback circuit converts a bank's own lending into its own market value, after accounting for all the value that leaks outward to other holders along the way.

Two features of this result deserve emphasis. First, the amplification is *nonlinear*—it grows convexly in both A_{ii} and Δ_t—so that heterogeneity in bank responses is most pronounced precisely when policy is most accommodative. Second, the result is silent on *which* banks have high A_{ii}. Since $\hat{C}_{ii}$ does not depend on τ, the dilution factor is a fixed feature of the ownership structure: core banks heavily held by others always have $\hat{C}_{ii}$ substantially below one. Whether the amplification factor can overcome this dilution depends on the network density and τ. In the frictionless benchmark ($\tau = 0$), the amplification factor operates at full force and can push A_{ii} above one for banks at the center of dense ownership loops. But in the empirically relevant regime ($\tau > 0$), the dilution factor dominates: core banks see much of their value flow outward through the network, leaving them with A_{ii} well below one, while peripheral banks with few incoming ownership links retain nearly all their value and have A_{ii} close to one. In this regime, Proposition 2 implies that *peripheral* banks are more elastic to policy—an intuitive result, since these banks' value feedback circuits are less attenuated by outward leakage.

Remark 2 (Role of τ)**.** *The friction parameter governs the cross-sectional dispersion of dependency indices through the amplification factor alone, leaving the dilution factor unchanged. At $\tau = 0$, the amplification factor operates at full force and A_{ii} can exceed one for banks at the center of dense ownership loops. As τ increases, the amplification factor is attenuated by $(1-\tau)^k$ at each path length k, progressively compressing the dispersion of A_{ii}. In the limit $\tau \to 1$, $\mathbf{A} \to \hat{\mathbf{C}}$ and the heterogeneous response mechanism is driven solely by ownership structure rather than by network feedback. The empirically relevant regime is intermediate: τ is large enough to prevent explosive feedback but small enough to generate economically significant heterogeneity beyond what ownership shares alone would produce.*

In general equilibrium, Proposition 2 provides the micro-foundation for two aggregate phenomena. First, the monotonicity $\partial \mathcal{Z}_{i,t}/\partial A_{ii} > 0$ helps generate credit reallocation across banks that carry different log-effective weights $\tilde{\omega}_j^{\ell}$. Second, inherited credit positions create persistent financial wedges in the aggregate system.

Policy-induced credit reallocation. We now examine how the heterogeneous bank responses identified in Proposition 2 reshape the aggregate market structure. Let aggregate credit supply be $B_t \equiv \sum_{i \in \mathcal{N}} B_{i,t}$, and define bank i's market share as $\theta_{i,t} \equiv B_{i,t}/B_t$. We measure concentration using the Herfindahl–Hirschman Index: $HHI_t \equiv \sum_{i \in \mathcal{N}} \theta_{i,t}^2$.

Definition 3 (Share-Weighted Moments). *For any cross-sectional variables $\{y_{i,t}\}$ and $\{z_{i,t}\}$, define the share-weighted expectation and covariance:*

$$\mathbb{E}_{\theta,t}[y_{i,t}] \equiv \sum_{i\in\mathcal{N}} \theta_{i,t}\, y_{i,t}, \tag{15}$$

$$\mathrm{Cov}_{\theta,t}(y_{i,t}, z_{i,t}) \equiv \sum_{i\in\mathcal{N}} \theta_{i,t}\big(y_{i,t} - \mathbb{E}_{\theta,t}[y_{i,t}]\big)\big(z_{i,t} - \mathbb{E}_{\theta,t}[z_{i,t}]\big). \tag{16}$$

Proposition 3 (Policy-Induced Credit Reallocation). *Accommodative monetary policy ($d\Delta_t > 0$) reshapes the cross-sectional distribution of credit. Market share flows toward banks whose dependency index exceeds the share-weighted average, and away from banks whose index falls below it. The net effect on market concentration is:*

$$dHHI_t = 2 \cdot \mathrm{Cov}_{\theta,t}(\theta_{i,t},\, \mathcal{Z}_{i,t}) \cdot d\Delta_t, \tag{17}$$

where $\mathcal{Z}_{i,t}$ is the semi-elasticity from Proposition 2. Since $\mathcal{Z}_{i,t}$ is strictly increasing in A_{ii}, concentration rises if and only if $\mathrm{Cov}_{\theta,t}(\theta_{i,t}, A_{ii}) > 0$—that is, if and only if larger banks tend to have higher dependency indices.

Proof. See Appendix A.3. □

Proposition 3 establishes that monetary policy is not distributionally neutral: it actively reshapes the industrial organization of the banking sector. An expansionary shock stimulates all banks, but those with higher dependency indices expand disproportionately. Whether this concentrates or disperses credit depends on which banks have high A_{ii}. When τ is small and the amplification factor dominates, core banks have the highest A_{ii}; if these core banks are also the largest lenders, easing *concentrates* credit. When τ is large enough that the dilution factor dominates, core banks have the *lowest* A_{ii}; easing then *disperses* credit toward peripheral institutions—a "credit democratization" effect driven by network frictions. In either case, the cross-sectional distribution of credit shifts endogenously with policy.

Corollary 4 (Market Share Reallocation). *In response to a marginal increase in the credit spread $d\Delta_t > 0$, bank i's market share grows ($d\theta_{i,t} > 0$) if and only if its semi-elasticity exceeds the share-weighted average:*

$$d\ln\theta_{i,t} = \big(\mathcal{Z}_{i,t} - \bar{\mathcal{Z}}_t\big)\, d\Delta_t, \tag{18}$$

where $\bar{\mathcal{Z}}_t \equiv \mathbb{E}_{\theta,t}[\mathcal{Z}_{j,t}]$. Since $\mathcal{Z}_{i,t}$ is increasing in A_{ii}, capital flows toward banks with higher dependency indices.

The key implication for general equilibrium is that the cross-sectional credit distribution is endogenous to monetary policy, making the vector $\hat{\mathbf{b}}_t^{FI}$ an independent macroeco-

nomic state variable. Aggregated through log-effective network weights $\tilde{\omega}_j^\ell$, this distribution determines the output-inflation trade-off: the financial architecture of the banking sector shapes the effectiveness of monetary policy through a channel entirely absent from representative-intermediary models.

Remark 3 (Credit Distribution as Endogenous Macroeconomic State Variable)**.** *Propositions 2 and 3 together establish that the cross-sectional credit distribution is an* endogenous macroeconomic state variable *whenever banks are connected through a network. This is the micro-foundation for the effective network weight: $\tilde{\omega}_j^\ell \neq \omega_j$ implies that the same total credit, distributed differently, produces different aggregate outcomes.*

Network topology and the amplification-reallocation decomposition. The preceding results show that the network affects monetary transmission through two distinct channels: *amplification* (all banks expand more due to value feedback) and *reallocation* (credit shifts across banks with different dependency indices). A natural question is whether these two effects always move together, or whether network architectures exist that activate one without the other. The answer depends on the symmetry of the network.

Proposition 5 (Symmetric vs. Asymmetric Networks)**.** *Consider the class of networks parameterized by a uniform cross-holding intensity $\delta \in [0, 1)$ and a topology matrix $\mathbf{G}$ with $G_{ii} = 0$ and $\sum_j G_{ij} = 1$ for all i, so that $\mathbf{C} = \delta\mathbf{G}$.*

1. Symmetric networks. *If $\mathbf{G}$ is vertex-transitive (each bank occupies the same network position), then $A_{ii} = A_{jj}$ for all i, j. The network amplifies aggregate credit supply but does not generate reallocation: $dHHI_t = 0$ for a common $\phi_{i,t} = \bar{\phi}$.*

2. Asymmetric networks. *If banks occupy non-equivalent network positions, then dependency indices differ generically. The network activates both the amplification and reallocation channels, with the direction of reallocation governed by $\mathrm{Cov}_{\theta,t}(\theta_{i,t}, A_{ii})$.*

Proof. See Appendix A.4. □

Proposition 5 provides the theoretical anchor for distinguishing the effects of network *density* (the scalar δ) from network *asymmetry* (the symmetry properties of $\mathbf{G}$). Both dense vertex-transitive networks (e.g., a complete graph) and dense asymmetric networks (e.g., a core-periphery structure) amplify aggregate credit, since higher δ raises the average feedback multiplier. But only asymmetric networks generate differential dependency indices across banks and thereby activate the reallocation mechanism.

To illustrate, consider $n = 5$ banks with uniform intensity $\delta = 0.15$ under four architectures. (i) *Complete graph*: $G_{ij} = \frac{1}{n-1}$ for $j \neq i$; vertex-transitive, so A_{ii} is identical for all banks. (ii) *Ring*: each bank holds only its neighbor; vertex-transitive, so A_{ii} is identical. (iii) *Star*: bank 1 (center) holds equal stakes in banks 2–5, and peripheral

banks hold stakes only in bank 1; not vertex-transitive, so $A_{11} \neq A_{ii}$ for $i \geq 2$. (iv) *Core-periphery*: banks 1–2 (core) hold each other and peripheral banks 3–5, while peripheral banks hold only core banks; not vertex-transitive, so core and peripheral banks have different A_{ii}. Under architectures (i)–(ii), Proposition 5(a) applies: accommodative policy amplifies lending uniformly and $dHHI_t = 0$. Under (iii)–(iv), Proposition 5(b) applies: amplification is heterogeneous and the credit distribution shifts.

This decomposition carries a sharp implication for general equilibrium: symmetric networks can only amplify the demand channel of monetary transmission, while the supply-side reallocation channel—operating through effective market share wedges and Phillips Curve flattening—requires network asymmetry.

Discussion and preview of general equilibrium. A retrospective examination of the four results above reveals a striking feature of their proofs: none invokes the specific functional form $\mathbf{A} = \hat{\mathbf{C}}(\mathbf{I} - (1-\tau)\mathbf{C})^{-1}$. Every proposition relies only on the following general properties of $\mathbf{A}$:

1. $A_{ij} \geq 0$ for all i, j (non-negative value transmission);
2. $A_{ii} \in (0, 1]$ with cross-sectional heterogeneity (differential self-retention);
3. column sums $\sum_i A_{ij} \leq 1$, with strict inequality when $\tau > 0$ (value dissipation);
4. the Neumann expansion $\sum_k [(1-\tau)\mathbf{C}]^k$ converges (stability).

The dependency matrix $\mathbf{A}$ is therefore a *sufficient statistic* for the banking network's effect on credit supply: any micro-foundation that generates a matrix with these four properties will produce identical analytical results. Equity cross-holdings are one such micro-foundation—chosen for their parsimony and the precedent of Elliott et al. (2014)—but they are not the only one. In practice, bank interdependence arises from multiple channels: interbank lending, common asset exposures (Allen et al. 2012), derivatives counterparty relationships, deposit market spillovers, information contagion, and fire-sale externalities. The key empirical question is not whether any single channel dominates, but whether the *aggregate* dependency matrix—the net effect of all channels—exhibits the four properties listed above.

A large empirical literature has estimated objects that are the reduced-form counterparts of our structural $\mathbf{A}$, confirming that it possesses all the required properties. Adrian and Brunnermeier (2016) measure systemic risk contributions through CoVaR and find that individual banks' VaR and ΔCoVaR are only weakly correlated in the cross section—direct evidence that effective network weights $\tilde{\omega}_j^{\ell}$ differ from market shares ω_j. Diebold and Yılmaz (2014) construct connectedness measures from variance decompositions and document that system-wide connectedness is quantitatively large (approximately 60% in

normal times, exceeding 90% during crises), with a pronounced core-periphery asymmetry. Billio et al. (2012) use Granger-causality networks to show that financial institutions have become increasingly interconnected, with banks and insurance companies at the center of the network. These estimates consistently confirm that the aggregate bank dependency structure is quantitatively significant, asymmetric with a core-periphery pattern, and time-varying with the monetary policy stance—precisely the properties on which our theory relies.

The partial-equilibrium analysis has established four results driven entirely by the properties of the dependency matrix: *value dissipation* (Proposition 1); *heterogeneous credit elasticities* (Proposition 2); *endogenous credit reallocation* (Proposition 3); and *topology decomposition* (Proposition 5). These results take asset prices, the stochastic discount factor, and the credit spread as exogenous. Two questions remain open for general equilibrium.

First, how does the endogenous credit reallocation feed back to aggregate prices, output, and inflation? When banks reallocate credit, this changes borrower consumption, which changes aggregate demand, which changes the pricing kernel that governs bank valuations—a loop that is absent in partial equilibrium.

Second, does the network affect only the *level* of monetary transmission, or also its *composition* between demand and supply effects? The answer, as we will show, is that the network generates a supply-side channel that is entirely absent from the standard representative-intermediary framework. The PE object A_{ii} is not the final macroeconomic weight; it identifies which banks respond most to a policy-induced movement in spreads and asset prices. Aggregation maps that response into *log-effective network weights* $\tilde{\omega}_j^\ell \equiv \sum_i \omega_i \mathcal{A}_{ij}$ that generically differ from market shares ω_j whenever $\mathcal{A} \neq \mathbf{I}$. The network creates a wedge between *who expands credit* and *how much that credit matters for the macroeconomy*. Section 4 will show that the macroeconomic consequences depend on the alignment between high-response banks and high-effective-weight banks. When policy pushes credit toward banks with large effective weights, borrower-saver redistribution lowers marginal costs for a given level of output; when it does not, network amplification need not translate into Phillips Curve flattening. This financial reallocation channel—running from network topology through credit reallocation to the slope of the Phillips Curve—is the paper's central contribution. Answering these questions requires closing the model with households and a monetary authority, to which we now turn.

2.2 Households

Following Sims et al. (2023), the household sector consists of two types of agents: a representative unconstrained household (patient) and a representative constrained household (impatient). The patient household is the standard optimizing agent in New Keynesian

models: it consumes, supplies labor, saves through short-term deposits, and owns equity claims on both production firms and financial intermediaries. The impatient household is less patient, does not supply labor, and finances its consumption by issuing long-term bonds. Let $z_N \in [0,1)$ denote the population share of impatient households. The log-linear aggregate equations use the borrower steady-state consumption share $z_C \equiv \bar{C}_B/\bar{Y}$, which need not equal z_N away from a normalization with equal per-capita steady-state consumption.

2.2.1 The Patient Household

The representative patient household has preferences over consumption $C_{P,t}$ and labor L_t. It discounts future utility flows by the subjective discount factor $\beta \in (0,1)$. The patient household's lifetime utility is given by:

$$\mathcal{V}_t^P = \max\ \mathbb{E}_t \sum_{j=0}^{\infty} \beta^j \left[\frac{C_{P,t+j}^{1-\sigma} - 1}{1-\sigma} - \psi \frac{L_{t+j}^{1+\chi}}{1+\chi} \right], \tag{19}$$

where $\sigma > 0$ is the inverse of the intertemporal elasticity of substitution, $\chi \geq 0$ is the inverse Frisch elasticity of labor supply, and $\psi > 0$ is a scaling parameter governing the disutility of labor.

The patient household has access to several financial instruments and ownership claims. It saves through one-period nominal deposits $S_{i,t}$ held with financial intermediary i. These deposits pay a gross nominal interest rate R_t^s between periods t and $t+1$. The patient household owns all equity in the production sector and receives aggregate dividends $\bar{D}_t^F$ from final goods producers, retailers, and wholesale firms. It also holds the outside equity claims on the banking sector: recall from Section 2.1.1 that the underlying ownership structure is captured by $\hat{C}_{ii} \in (0,1]$, representing the fraction of bank i's equity held by households. The patient household receives dividends $P_t D_t^{P,FI}$ from this outside equity position. Besides, it makes a transfer X_t^b to the impatient household each period, as well as a transfer $P_t X_{i,t}^{FI}$ to each financial intermediary. Lastly, the patient household receives a lump-sum transfer T_t from the fiscal authority, which includes the central bank's operating surplus remitted to the treasury.

Treating all prices, dividends from firms, government transfers, the transfer schedule to the impatient household, and the equity injection schedule to banks as given, the representative patient household maximizes Equation (19) by choosing a sequence of consumption $C_{P,t}$, labor supply L_t, and one-period deposits $\{S_{i,t}\}_{i\in\mathcal{N}}$, subject to the budget constraint

$$P_t C_{P,t} + \sum_{i\in\mathcal{N}} S_{i,t} + \sum_{i\in\mathcal{N}} P_t X_{i,t}^{FI} + P_t X_t^b \leq W_t L_t + \sum_{i\in\mathcal{N}} R_{t-1}^s S_{i,t-1} + P_t D_t^F + P_t D_t^{P,FI} + P_t T_t. \tag{20}$$

The patient household's nominal stochastic discount factor (SDF), which prices nominal payoffs from the perspective of the household sector, is defined as:

$$M^N_{t,t+1} \equiv \beta \left(\frac{C_{P,t+1}}{C_{P,t}} \right)^{-\sigma} \frac{1}{\Pi_{t+1}}. \tag{21}$$

This SDF determines nominal asset prices, including the valuation of banks' bond portfolios and the equilibrium credit spread that were taken as exogenous in Section 2.1.3.

2.2.2 The Impatient Household

The representative impatient household derives utility solely from consumption $C_{B,t}$ and does not supply labor. The impatient household is less patient than the patient household, discounting future utility flows by $\beta_B < \beta$. This relative impatience generates a borrowing motive that sustains the demand for long-term credit in equilibrium. The impatient household's lifetime utility is given by:

$$\mathcal{V}^B_t = \max \ \mathbb{E}_t \sum_{j=0}^{\infty} \beta_B^j \left[\frac{(C_{B,t+j})^{1-\sigma} - 1}{1-\sigma} \right]. \tag{22}$$

We assume the same intertemporal elasticity of substitution $1/\sigma$ for both household types, which simplifies aggregation while preserving the essential heterogeneity in patience that drives credit demand.

Each period, the impatient household must finance consumption expenditure $P_t C_{B,t}$ plus coupon payments on outstanding debt B^C_{t-1} using proceeds from new bond issuance and the transfer received from the patient household. The impatient household's nominal budget constraint is:

$$P_t C_{B,t} + B^C_{t-1} \leq Q_t \left(B^C_t - \kappa B^C_{t-1} \right) + P_t X^b_t, \tag{23}$$

where the left-hand side represents total uses of funds (consumption plus debt service) and the right-hand side represents total sources (new borrowing plus the transfer from the patient household).

We assume that the transfer from the patient to the impatient household is set according to

$$P_t X^b_t = (1 + \kappa Q_t) B^C_{t-1}, \tag{24}$$

which implies that the patient household effectively pays off the impatient household's maturing debt obligations each period. Substituting into the budget constraint (23) and simplifying, we obtain that borrower consumption is determined by a static relationship:

$$P_t C_{B,t} = Q_t B^C_t. \tag{25}$$

The borrower's real consumption therefore equals the real market value of its outstanding debt.

This transfer rule serves as a simplifying device that eliminates B^C_{t-1} as a relevant state variable for the impatient household's optimization, reducing the dimensionality of the system. It ensures that borrower consumption dynamics are driven entirely by current credit conditions rather than by the history of debt accumulation, allowing us to isolate the effects of financial frictions from intertemporal debt dynamics.

Although the constraint appears static, the impatient household's Euler equation involves expectations of future bond returns because current bond issuance B^C_t determines next period's coupon obligations, which affect the terms at which debt can be rolled over. To ensure stationarity of the debt position, we introduce a small portfolio holding cost following Schmitt-Grohé and Uribe (2003): the impatient household's optimality condition takes the form

$$1 = \beta_B \left(\frac{C_{B,t+1}}{C_{B,t}} \right)^{-\sigma} \frac{R_{b,t+1}}{\Pi_{t+1}} - \psi_b \left(\frac{b_{C,t}}{\bar{b}_C} - 1 \right),$$

where $\psi_b \geq 0$ is a small debt-elasticity parameter. This term penalizes deviations of borrower debt from its steady-state level, preventing the near-unit-root dynamics that arise in incomplete-markets models. The term vanishes in steady state ($b_{C,t} = \bar{b}_C$) and is kept in the bond-pricing closure rather than appended directly to the benchmark IS curve in Section 3.1.

The impatient household plays a crucial role in generating a meaningful demand for long-term credit intermediated by the banking sector. In the absence of such demand, financial intermediaries would have no profitable lending opportunities, and the dependency network would be irrelevant for equilibrium dynamics. The impatient household's relative impatience ($\beta_B < \beta$) ensures that in steady state it is a net borrower while the patient household is a net saver, with financial intermediaries channeling funds between the two household types. From an economic interpretation perspective, the impatient household can be viewed as a reduced-form representation of the investment sector or the market for durable consumption goods; the dynamics of borrower consumption in this framework align closely with the behavior of investment in richer models with physical capital accumulation (Sims and Wu 2021).

2.3 Central Bank and Monetary Policy

The central bank conducts monetary policy through two instruments: a short-term policy rate and a portfolio of long-term bonds. This dual-instrument framework allows us to study both conventional interest rate policy and unconventional balance sheet policies (quantitative easing).

Central bank balance sheet. The central bank holds a portfolio of long-term bonds B_t^{cb} issued by the impatient household, financed by the creation of reserves RE_t. The central bank's balance sheet identity is:

$$Q_t B_t^{cb} = RE_t. \tag{26}$$

We define the real value of the central bank's bond portfolio as $QE_t \equiv Q_t B_t^{cb}/P_t = q_t b_t^{cb}$, where $q_t \equiv Q_t/P_t$ is the real bond price and $b_t^{cb} \equiv B_t^{cb}/P_t$ is real bond holdings. The central bank can freely adjust its bond portfolio, which constitutes the quantitative easing (QE) instrument.

Policy rules. The short-term nominal interest rate R_t^s follows a Taylor-type rule that responds to deviations of inflation and output from their targets:

$$\frac{R_t^s}{R^s} = \left(\frac{R_{t-1}^s}{R^s}\right)^{\rho_r} \left[\left(\frac{\Pi_t}{\Pi^*}\right)^{\phi_\pi} \left(\frac{Y_t}{Y_t^*}\right)^{\phi_y}\right]^{1-\rho_r} e^{\varepsilon_{r,t}}, \tag{27}$$

where R^s is the steady-state nominal interest rate, Π^* is the inflation target, Y_t^* is potential output, $\rho_r \in [0,1)$ captures interest rate smoothing, $\phi_\pi > 1$ and $\phi_y \geq 0$ are the response coefficients to inflation and the output gap, and $\varepsilon_{r,t}$ is a monetary policy shock.

The central bank's bond portfolio follows a similar rule:

$$\frac{QE_t}{QE} = \left(\frac{QE_{t-1}}{QE}\right)^{\rho_q} \left[\left(\frac{\Pi_t}{\Pi^*}\right)^{-\lambda_\pi} \left(\frac{Y_t}{Y_t^*}\right)^{-\lambda_y}\right]^{1-\rho_q} e^{\varepsilon_{q,t}}, \tag{28}$$

where QE is the steady-state bond portfolio, $\rho_q \in [0,1)$ is the smoothing parameter, $\lambda_\pi \geq 0$ and $\lambda_y \geq 0$ govern the response to inflation and output, and $\varepsilon_{q,t}$ is a QE shock. The negative signs reflect that the central bank expands its bond portfolio (easing) when inflation or output fall below target.

Interest on reserves. The central bank pays interest R_t^{re} on reserves held by financial intermediaries. We assume that the central bank sets the interest on reserves equal to the short-term policy rate:

$$R_t^{re} = R_t^s. \tag{29}$$

This assumption, combined with the no-arbitrage condition from the financial intermediaries' problem, implies that banks are indifferent between holding reserves and short-term deposits, and that reserves yield no excess return in expectation.

Central bank surplus and fiscal transfer. The central bank earns returns on its bond portfolio and pays interest on reserves. The operating surplus (or deficit) is remitted

to the fiscal authority, which then transfers it to the patient household. This transfer satisfies:

$$P_t T_t = R_t^b Q_{t-1} B_{t-1}^{cb} - R_{t-1}^{re} RE_{t-1}, \tag{30}$$

where $R_t^b Q_{t-1} B_{t-1}^{cb}$ is the return on the central bank's bond portfolio (coupon plus capital gain) and $R_{t-1}^{re} RE_{t-1}$ is the interest paid on reserves. When the surplus is negative, the transfer becomes a lump-sum tax on the patient household. This specification ensures that the government budget constraint is satisfied and that central bank operations do not create fiscal imbalances.

3 Aggregation and the Role of Effective Network Weights

We now log-linearize the general equilibrium model and reduce it to a compact set of four structural equations. Throughout, hatted variables $\hat{x}_t \equiv \ln(x_t/\bar{x})$ denote log-deviations from steady state. The key distinction is that the level dependency matrix $\mathbf{A}$ is not itself a log-linear elasticity. The log-linearization derivations are collected in Appendix C.

3.1 The Aggregate Demand Side (IS Curve)

Define the *net financial capacity* of the banking sector as

$$\Theta_t^{\text{net}} \equiv \hat{q}_t + \hat{b}_t^{FI}, \tag{31}$$

which combines the real bond price and aggregate bank lending into a single sufficient statistic for the banking sector's contribution to aggregate demand. Define also the real QE variable $\widehat{qe}_t \equiv \hat{q}_t + \hat{b}_t^{cb}$. The same bond-price movement raises the market value of both private and central-bank bond absorption, so the two capacity measures share $\hat{q}_t$ but keep the holders separate.

Log-linearizing goods market clearing, borrower consumption, and the patient household's Euler equation yields:

$$\hat{y}_t = \mathbb{E}_t \hat{y}_{t+1} - \frac{1 - z_C}{\sigma}\left(\hat{r}_t^s - \mathbb{E}_t \hat{\pi}_{t+1}\right) + z_C \bar{b}^{FI}\left(\Theta_t^{\text{net}} - \mathbb{E}_t \Theta_{t+1}^{\text{net}}\right) + z_C \bar{b}^{CB}\left(\widehat{qe}_t - \mathbb{E}_t \widehat{qe}_{t+1}\right), \tag{32}$$

where $z_C \equiv \bar{C}_B/\bar{Y}$ is the borrower's steady-state consumption share, and $\bar{b}^{FI}$ and $\bar{b}^{CB}$ are the steady-state shares of borrower debt held by intermediaries and the central bank. The borrower debt-elastic closing device affects the bond-pricing block; it is not appended directly to the IS curve. Financial capacity enters in changes because borrower spending responds to current credit conditions relative to expected future conditions. A purely permanent increase in capacity has less intertemporal bite than a temporary one.

3.2 The Aggregate Supply Side (NKPC)

With Calvo pricing frictions, the augmented Phillips Curve is:

$$\hat{\pi}_t = \beta\,\mathbb{E}_t\hat{\pi}_{t+1} + \kappa_y\,\hat{y}_t - \Lambda_C\big(\bar{b}^{FI}\,\Theta_t^{\text{net}} + \bar{b}^{CB}\,\widehat{qe}_t\big) - \psi_a\,\hat{a}_t, \tag{33}$$

where

$$\kappa_y \equiv \lambda_p\left(\chi + \frac{\sigma}{1-z_C}\right), \quad \Lambda_C \equiv \lambda_p\,\frac{\sigma z_C}{1-z_C}, \quad \psi_a \equiv \lambda_p(1+\chi), \quad \lambda_p \equiv \frac{(1-\theta_p)(1-\theta_p\beta)}{\theta_p}. \tag{34}$$

κ_y is the usual output slope after borrower-saver aggregation, Λ_C is the strength of the credit-induced marginal-cost offset, and ψ_a is the pass-through from productivity to marginal cost. The parameter θ_p is the Calvo non-adjustment probability and is distinct from the bank portfolio adjustment cost η_B. The new term is the supply-side counterpart of credit expansion: higher borrower consumption lowers saver consumption for a given output level, which reduces marginal cost through the saver's labor supply condition.

3.3 The Credit Distribution and the Law of Motion

Define

$$s_j \equiv \frac{\bar{\Delta}\,\bar{q}\,\bar{B}_j}{\bar{x}_j}, \qquad e_j \equiv \frac{\kappa\,\bar{q}\,\bar{B}_j}{\bar{x}_j}, \tag{35}$$

where s_j is the share of bank j's steady-state fundamental value attributable to lending operations and e_j captures the sensitivity to the inherited bond portfolio. Let $\mathbf{S} \equiv \text{diag}(s_1,\ldots,s_n)$ and $\mathbf{E} \equiv \text{diag}(e_1,\ldots,e_n)$. Thus s_j loads current spread income, while e_j loads the legacy portfolio channel that makes past lending matter today.

Substituting the linearized fundamental values into the leverage constraints gives:

$$\hat{\mathbf{b}}_t^{FI} := \underbrace{\big[(1+\eta_B)\mathbf{I} - \mathcal{A}\mathbf{S}\big]^{-1}}_{\text{Damped Credit Multiplier}}\Big[\hat{\boldsymbol{\phi}}_t + \boldsymbol{\alpha}_q\,\hat{q}_t + \boldsymbol{\alpha}_\Delta\,\hat{\Delta}_t + \underbrace{\big(\mathcal{A}\mathbf{E} + \eta_B\mathbf{I}\big)}_{\text{Augmented Legacy}}\,\hat{\mathbf{b}}_{t-1}^{FI}\Big], \tag{36}$$

The inverse is a damped credit multiplier: network valuation feedback raises lending capacity, while η_B limits how sharply portfolios can adjust. The bracketed terms are the current shocks, asset-price and spread movements, and inherited balance-sheet positions that feed into lending. where

$$\boldsymbol{\alpha}_q \equiv \mathcal{A}(\mathbf{S}+\mathbf{E})\mathbf{1} - \mathbf{1}, \qquad \boldsymbol{\alpha}_\Delta \equiv \mathcal{A}\mathbf{S}\mathbf{1}. \tag{37}$$

$\boldsymbol{\alpha}_q$ measures how bond prices affect lending after both current and legacy valuation effects, while $\boldsymbol{\alpha}_\Delta$ isolates the spread-income channel. In both cases, $\mathcal{A}$ converts bank-level fundamentals into market values before leverage constraints bind. The inverse exists

under the matrix stability condition $\rho((1+\eta_B)^{-1}\mathcal{A}\mathbf{S}) < 1$. This condition rules out an explosive contemporaneous loop from lending to valuation and back to lending.

The persistence of the credit distribution is governed by

$$\mathbf{P} \equiv \big[(1+\eta_B)\mathbf{I} - \mathcal{A}\mathbf{S}\big]^{-1}\big(\mathcal{A}\mathbf{E} + \eta_B\mathbf{I}\big). \tag{38}$$

This matrix summarizes financial memory: inherited portfolios affect current values directly through $\mathbf{E}$ and mechanically through the portfolio adjustment term.

3.4 Network Aggregation of Financial Capacity

Weighting the linearized leverage constraints by steady-state lending shares $\omega_i \equiv \bar{q}\bar{B}_i / \sum_j \bar{q}\bar{B}_j$ yields:

$$\Theta_t^{\text{net}} := \hat{\phi}_t^{agg} + \bar{\mu}\,\hat{\Delta}_t + \bar{\alpha}\,\hat{q}_t + \mathbf{\Lambda}^{(0)}\hat{\mathbf{b}}_t^{FI} + \mathbf{\Lambda}^{(1)}\hat{\mathbf{b}}_{t-1}^{FI} - \eta_B\big(\hat{b}_t^{FI} - \hat{b}_{t-1}^{FI}\big), \tag{39}$$

where $\hat{\phi}_t^{agg} \equiv \sum_i \omega_i \hat{\phi}_{i,t}$ and $\hat{b}_t^{FI} \equiv \sum_i \omega_i \hat{b}_{i,t}$. This equation turns the bank-level credit distribution into a scalar financial-capacity state. Spreads, bond prices, and credit at high-weight banks raise capacity, while rapid aggregate portfolio adjustment absorbs resources. The aggregation coefficients are determined by the log-linear effective network weight:

$$\tilde{\omega}_j^{\ell} \equiv \sum_{i\in\mathcal{N}} \omega_i\,\mathcal{A}_{ij}. \tag{40}$$

The effective weight asks how much bank j's fundamental value moves the market-value base of all lending-weighted banks, not how large bank j is on its own balance sheet. The coefficients in (39) are:

$$\bar{\mu} = \sum_j \tilde{\omega}_j^{\ell}\,s_j, \quad \bar{\alpha} = \sum_j \tilde{\omega}_j^{\ell}\,(s_j + e_j), \quad \Lambda_j^{(0)} = \tilde{\omega}_j^{\ell}\,s_j, \quad \Lambda_j^{(1)} = \tilde{\omega}_j^{\ell}\,e_j. \tag{41}$$

Each coefficient is an effective network weight multiplied by the relevant source of value exposure. Current spread and price effects load on s_j, while inherited positions load on e_j.

The level dependency matrix $\mathbf{A}$ still gives the structural network object. The effective weight $\tilde{\omega}_j^{\ell}$ is its log-linear aggregation counterpart: it measures how bank j's log fundamental value contributes to aggregate financial capacity after accounting for steady-state valuation shares.

3.5 The Credit Spread

The credit spread is defined with the nominal SDF:

$$\Delta_t \equiv \mathbb{E}_t[M_{t,t+1}^N(R_{t+1}^b - R_t^s)]. \tag{42}$$

This is the discounted excess return on bank lending over deposits, so it is the profitability margin that enters bank fundamental value. Let $\mathcal{R} \equiv \beta/\beta_B > 1$, so $\bar{\Delta} = \mathcal{R} - 1$. Log-linearizing the discounted spread and using the saver Euler equation $\mathbb{E}_t[M^N_{t,t+1} R^s_t] = 1$ gives:

$$\hat{\Delta}_t = \frac{\mathcal{R}}{\mathcal{R} - 1}\big(\mathbb{E}_t \hat{r}^b_{t+1} - \hat{r}^s_t\big). \tag{43}$$

The spread rises when expected long-bond returns increase relative to the deposit rate, scaled by the steady-state excess-return margin. The long bond return linearizes as:

$$\hat{r}^b_{t+1} \approx \beta_B \kappa\, \hat{q}_{t+1} - \hat{q}_t, \tag{44}$$

where $\beta_B \kappa = \kappa \bar{q}/(1 + \kappa \bar{q})$ under the steady-state borrower pricing condition. A higher price today lowers the holding-period return, while a higher expected price tomorrow raises the continuation value of the long bond.

4 Macroeconomic Consequences of the Effective Network Weight

The partial-equilibrium propositions in Section 2 identify the response margin: which banks expand lending after a policy shock. The general-equilibrium system derived in Section 3 identifies the aggregation margin: whether those bank-level responses move aggregate financial capacity. This section connects the two margins. The dependency index A_{ii} governs how strongly bank i's own lending feeds back into its own market value and relaxes its leverage constraint. The effective network weight $\tilde{\omega}^\ell_j$ governs how much a change in bank j's fundamental value moves the financial capacity of the banking system as a whole.

The distinction matters because the banks that respond most to monetary easing need not be the banks whose lending matters most for the macroeconomy. Proposition 2 and Proposition 3 show how policy changes the credit distribution through heterogeneous bank responses. The four-equation system shows how that distribution, once weighted by $\tilde{\omega}^\ell_j$, changes borrower demand, saver labor supply, and inflation. The central macroeconomic condition is therefore an alignment condition: monetary transmission has the largest aggregate capacity effect when policy pushes credit toward banks that are both highly responsive and highly effective in the network.

4.1 From Response Indices to Macro Weights

The PE and GE objects answer different economic questions. The dependency index A_{ii} asks how much of bank i's own fundamental value is retained in bank i's own market

value after all ownership feedback and value leakage are accounted for. It is therefore the object that governs bank i's lending elasticity in Proposition 2. The market share ω_j asks how much credit bank j directly holds in steady state. The effective network weight asks a third question: when bank j's fundamental value changes, how much does the market-value base of the entire banking sector move?

The answer is

$$\tilde{\omega}_j^{\ell} \equiv \sum_i \omega_i \mathcal{A}_{ij}. \tag{45}$$

The term $\mathcal{A}_{ij}$ says how much bank j's log fundamental value changes bank i's log market value; the weight ω_i says how much bank i matters in aggregate lending capacity. A bank can therefore be highly responsive without being macroeconomically central, and it can be macroeconomically central without being the most responsive bank.

This is where the partial-equilibrium results enter the macroeconomic analysis. PE determines the direction of credit reallocation across banks; GE determines how much that reallocation changes Θ_t^{net}. In a representative-intermediary model, the question "which bank holds the credit" has no aggregate content: only the total quantity matters. Here the same aggregate stock of credit can generate different values of Θ_t^{net} depending on whether the marginal lending is carried by banks with high or low $\tilde{\omega}_j^{\ell}$.

4.2 The Transmission Mechanism

The first step is the bank-level response. An expansionary monetary shock raises the value of long bonds, lowers funding costs, or changes the expected spread between bank assets and deposits. Proposition 2 shows that banks do not respond uniformly: the same movement in prices or spreads creates larger lending responses at banks with stronger self-feedback through A_{ii}. A bank that expands its bond portfolio raises its own fundamental value through spread income and through the value of inherited bond positions; the network then maps that fundamental gain into market values and leverage capacity.

The second step is reallocation. Because banks have different response indices, the credit distribution $\hat{\mathbf{b}}_t^{FI}$ changes endogenously. Proposition 3 formalizes this point: credit flows toward banks whose semi-elasticity exceeds the share-weighted average. This step says where marginal credit goes. It does not yet say whether the reallocation is macroeconomically powerful.

The third step is topology. Proposition 5 separates uniform amplification from genuine reallocation. A vertex-transitive network can raise lending through a common feedback multiplier without changing the credit distribution. An asymmetric network generates different dependency indices, so easing can move credit across banks with different network positions. Only the second case activates the composition channel: policy changes not only how much credit expands, but also which banks supply the marginal credit and

how strongly that credit maps into aggregate financial capacity.

The fourth step is GE aggregation. A dollar of lending at bank j changes bank j's balance sheet, but it also changes the market values of banks that hold claims on j or are indirectly exposed to it. Those valuation changes feed back into leverage constraints and alter the system's ability to support credit. The network therefore converts a cross-sectional allocation into an aggregate object, Θ_t^{net}, only after the reallocation is weighted by $\tilde{\omega}_j^{\ell}$.

The final step is household redistribution. Borrower consumption is tied to the real value of debt financed by intermediaries and the central bank. Given output, an increase in borrower consumption lowers saver consumption:

$$\hat{c}_{P,t} = \frac{1}{1 - z_C}\hat{y}_t - \frac{z_C}{1 - z_C}\big(\bar{b}^{FI}\Theta_t^{\text{net}} + \bar{b}^{CB}\widehat{qe}_t\big). \tag{46}$$

The fall in saver consumption raises the saver's marginal utility of income. Through the labor supply condition, this shifts effective labor supply outward and lowers real marginal cost for a given level of output. The network therefore changes the output–inflation composition of monetary transmission through the composition of credit, not only through the level of aggregate demand.

4.3 The Dual Channel: Demand and Marginal Cost

The same financial-capacity variable enters the two core macroeconomic equations in different ways. In the IS curve it enters in changes:

$$\hat{y}_t = \mathbb{E}_t\hat{y}_{t+1} - \frac{1 - z_C}{\sigma}\big(\hat{r}_t^s - \mathbb{E}_t\hat{\pi}_{t+1}\big) + z_C\bar{b}^{FI}\big(\Theta_t^{\text{net}} - \mathbb{E}_t\Theta_{t+1}^{\text{net}}\big) + z_C\bar{b}^{CB}\big(\widehat{qe}_t - \mathbb{E}_t\widehat{qe}_{t+1}\big). \tag{47}$$

The reason is intertemporal. Borrower consumption rises when current financial capacity is high relative to expected future financial capacity. A temporary easing that raises current bank lending or current QE purchases stimulates current borrower spending. If the same increase is expected to persist one-for-one into the future, the intertemporal impulse to current demand is weaker.

In the Phillips Curve, financial capacity enters in levels:

$$\hat{\pi}_t = \beta\mathbb{E}_t\hat{\pi}_{t+1} + \kappa_y\hat{y}_t - \Lambda_C\big(\bar{b}^{FI}\Theta_t^{\text{net}} + \bar{b}^{CB}\widehat{qe}_t\big) - \psi_a\hat{a}_t. \tag{48}$$

Inflation depends on today's marginal cost. For a given output level, a higher level of borrower consumption implies lower saver consumption today. Lower saver consumption increases the marginal value of income for the household that supplies labor, which lowers the real wage required to support a given amount of labor. Marginal cost falls today even if the movement in financial capacity is expected to persist.

This is the output–inflation composition that the network changes. A standard financial accelerator mainly says that easing raises credit and therefore aggregate demand. Here, the network also determines how much of that credit expansion shows up as a supply-side offset. The strength of the offset relative to the direct output term is

$$\frac{\Lambda_C}{\kappa_y} = \frac{\sigma z_C}{\chi(1 - z_C) + \sigma}. \tag{49}$$

The term z_C controls how much credit-financed borrower spending matters in goods-market clearing. The curvature parameter σ controls how strongly saver marginal utility reacts when saver consumption falls. The network does not change these preference parameters. It changes the financial-capacity movement to which these parameters are applied.

4.4 Output–Inflation Composition

To see how the network changes the output–inflation composition of monetary transmission, substitute the credit distribution law into the aggregation equation. The contemporaneous transmission from bank-level shocks and prices into aggregate capacity is summarized by

$$\boldsymbol{\Gamma}' \equiv \boldsymbol{\Lambda}^{(0)}[(1 + \eta_B)\mathbf{I} - \mathcal{A}\mathbf{S}]^{-1}. \tag{50}$$

This row vector combines the two margins emphasized above. The vector $\boldsymbol{\Lambda}^{(0)}$ identifies where effective network weights are large in aggregation. The inverse matrix identifies how shocks, spreads, and asset prices propagate through bank-level credit positions. A bank matters most for aggregate capacity when it is both important in aggregation and connected to propagation paths that make its lending response large.

The reduced-form financial capacity equation is:

$$\begin{aligned}\Theta_t^{\text{net}} = \hat{\phi}_t^{agg} + \boldsymbol{\Gamma}'\hat{\boldsymbol{\phi}}_t + \underbrace{\left(\bar{\mu} + \boldsymbol{\Gamma}'\boldsymbol{\alpha}_\Delta\right)}_{\equiv \mu^*} \hat{\Delta}_t + \underbrace{\left(\bar{\alpha} + \boldsymbol{\Gamma}'\boldsymbol{\alpha}_q\right)}_{\equiv \alpha^*} \hat{q}_t \\ + \underbrace{\left[\boldsymbol{\Lambda}^{(1)} + \boldsymbol{\Gamma}'(\mathcal{A}\mathbf{E} + \eta_B\mathbf{I})\right]}_{\equiv \boldsymbol{\Psi}'} \hat{\mathbf{b}}_{t-1}^{FI} - \eta_B(\hat{b}_t^{FI} - \hat{b}_{t-1}^{FI}).\end{aligned} \tag{51}$$

The coefficients μ^* and α^* show how spreads and bond prices move aggregate financial capacity after bank-level responses are network-weighted. The vector $\boldsymbol{\Psi}'$ shows how inherited credit positions enter today's capacity. The adjustment-cost term subtracts rapid aggregate portfolio changes, so a network can contain high effective-weight banks and still generate a muted aggregate response when portfolio adjustment is sluggish.

For a monetary disturbance, define

$$\gamma_t \equiv \left.\frac{\partial \Theta_t^{\text{net}}}{\partial \hat{y}_t}\right|_{\text{monetary shock}}. \tag{52}$$

This derivative measures how much financial capacity moves together with output under the policy experiment being considered. Because the derivative is taken along a particular monetary experiment, γ_t is not a primitive parameter; it is a reduced-form sufficient statistic for how that experiment moves financial capacity together with output. Substituting this co-movement into the Phillips Curve gives an equivalent representation:

$$\hat{\pi}_t = \beta \mathbb{E}_t \hat{\pi}_{t+1} + \tilde{\kappa}_t \hat{y}_t + \tilde{u}_t, \qquad \tilde{\kappa}_t \equiv \kappa_y - \Lambda_C \bar{b}^{FI} \gamma_t. \tag{53}$$

The textbook coefficient κ_y measures how inflation responds to output holding the financial redistribution term fixed. The second term captures the marginal-cost relief generated by the financial-capacity movement that accompanies monetary easing. We use this representation only to summarize the output–inflation composition of a policy experiment: a larger positive co-movement between output and network financial capacity means that a given output response is accompanied by a larger supply-side credit offset.

Proposition 6 (Financial Reallocation and Output–Inflation Composition)**.** *For monetary shocks with $\gamma_t > 0$, an increase in network financial capacity creates a marginal-cost offset to the inflationary effect of higher output. Relative to a no-network economy, monetary transmission shifts further toward real activity, and away from inflation, if and only if $\gamma_t^{net} > \gamma_t^{no\,net}$. This condition is more likely when banks with high response elasticities also have large effective network weights in aggregation.*

The condition in the proposition is the GE version of the PE reallocation results. Proposition 2 identifies the banks that respond most to easing, and Proposition 3 tells us whether credit moves toward them. The composition result asks a second question: do those high-response banks also have large aggregate-capacity effects? When the answer is yes, the same credit expansion generates a larger increase in Θ_t^{net}, a stronger redistribution from savers to borrowers, and a larger marginal-cost offset. Monetary policy then produces more real activity relative to inflation.

The condition can fail for economically informative reasons. If value dissipation is strong, if the banks expanding most have low effective weights, or if η_B makes the portfolio response sluggish, the network may add little to γ_t. A symmetric network illustrates the same point. Proposition 5 implies that a vertex-transitive network can amplify all banks uniformly without generating credit reallocation. In that case the network may change the level of financial transmission, but the output–inflation composition channel is muted. An asymmetric network matters for composition only when it moves credit across banks with different effective network weights.

4.5 Asymmetric Transmission of Policy Instruments

The same PE-GE logic explains why different policy instruments can be amplified differently. The partial-equilibrium propositions apply to the component of policy that changes bank balance sheets, spreads, and bond portfolios. They do not apply to the component that reaches households directly through the Euler equation. The network therefore amplifies the bank-balance-sheet share of a policy instrument, not the entire policy instrument mechanically.

Conventional interest rate policy has both components. A cut in $\hat{r}_t^s$ raises saver demand through the Euler equation. This is the standard New Keynesian channel and does not depend on which banks hold credit. The same rate cut also changes spreads and long-bond prices, which affects bank lending and therefore Θ_t^{net}. Only this second component is filtered through $\tilde{\omega}_j^\ell$ and the credit multiplier.

QE is more concentrated in the balance-sheet component. A purchase of long bonds changes the quantity of bonds held outside the central bank, raises bond prices, and changes the value of portfolios inherited by new bankers. These movements operate through the coefficients $\boldsymbol{\alpha}_q$, $\mathbf{E}$, and the aggregation weights in the financial block. QE therefore loads more heavily on the same feedback-and-reallocation mechanism identified by Proposition 2 and Proposition 3.

Proposition 7 (Conditional Asymmetric Network Amplification)**.** *Suppose the network multiplier raises the balance-sheet component of both policy instruments by a common factor greater than one, and suppose the interest-rate instrument has a positive direct Euler-equation component that is invariant to the network. Then the output amplification ratio for QE exceeds the output amplification ratio for conventional interest-rate policy.*

The comparison is about ratios, not only levels. If the network increases the balance-sheet response by the same proportional amount for both instruments, the instrument with the larger balance-sheet share receives the larger overall network amplification. QE has a larger such share because its transmission is routed through bond markets and intermediary balance sheets. Rate policy has a household bypass, so part of its output effect is present even when the network is shut down.

This also changes the output–inflation composition of the two instruments. A policy tool that produces more of its output response through Θ_t^{net} also produces more of the marginal-cost offset described above. QE can therefore generate a larger real response with a comparatively muted inflation response when its balance-sheet channel is strong. The claim does not require QE to dominate rate policy in every calibration. It requires the network-amplified share of QE transmission to be larger.

4.6 Inherited Financial Wedges

The financial reallocation channel also has memory. Monetary policy changes the cross-sectional allocation of bank credit, and that allocation persists because banks enter subsequent periods with legacy bond portfolios. These inherited portfolios enter the equity transfer to new bankers and affect current market values, current leverage constraints, and current network financial capacity. The first-order system summarizes this persistence through the matrix

$$\mathbf{P} = [(1+\eta_B)\mathbf{I} - \mathcal{A}\mathbf{S}]^{-1}(\mathcal{A}\mathbf{E} + \eta_B\mathbf{I}). \tag{54}$$

When $\rho(\mathbf{P})$ is close to one, credit reallocations fade slowly. Iterating the credit law forward shows that the effect of inherited positions decays with powers of $\mathbf{P}$. A period of easing that tilts credit toward a particular set of banks therefore leaves a trace in the next period's financial capacity. The term $\mathbf{\Psi}'\hat{\mathbf{b}}^{FI}_{t-1}$ is the macroeconomic expression of this inherited balance-sheet channel. It shows that the same aggregate stock of bonds can support different financial capacity depending on which banks hold it.

In the first-order system, $\mathcal{A}$ and $\tilde{\omega}^{\ell}$ are fixed around the steady state. Financial memory therefore operates through inherited wedges rather than through time-varying local coefficients. From (51), inherited credit positions enter through $\mathbf{\Psi}'\hat{\mathbf{b}}^{FI}_{t-1}$. Substituting this term into the NKPC shifts the inflation residual:

$$-\Lambda_C \bar{b}^{FI} \mathbf{\Psi}'\hat{\mathbf{b}}^{FI}_{t-1}.$$

Past policy changes the financial position from which the next policy shock is transmitted, even when the local coefficients are fixed.

Proposition 8 (Financial Memory Through Inherited Bond Holdings)**.** *In the first-order system with fixed $\mathcal{A}$ and $\tilde{\omega}^{\ell}$, the inherited cross-section of bank bond holdings affects current network financial capacity through $\mathbf{\Psi}'\hat{\mathbf{b}}^{FI}_{t-1}$ and evolves according to the persistence matrix $\mathbf{P}$. The persistence of this inherited financial wedge is increasing in $\rho(\mathbf{P})$.*

This is a form of financial memory. If past easing leaves the banking sector tilted toward high-effective-weight institutions, the economy enters the next period with a larger financial-capacity wedge. If past tightening shifts credit away from those institutions, the wedge moves in the opposite direction. The linear benchmark keeps the distinction clear: history shifts the inherited financial wedge, while richer state-dependent versions would allow market shares or effective network weights themselves to evolve over time.

4.7 Nesting, Non-Nesting, and Irreducibility

The degenerate network case clarifies what the network adds and what it does not add. The borrower-saver structure already creates a supply-side credit wedge: when borrower

consumption rises and saver consumption falls, marginal cost declines. That mechanism does not require a network. The network makes the wedge depend on the composition of intermediary credit.

Proposition 9 (Degenerate Network Limit)**.** *When* $\mathbf{C} = \mathbf{0}$*, the level dependency matrix is* $\mathbf{A} = \mathbf{I}$ *and the log-elasticity matrix is also* $\mathcal{A} = \mathbf{I}$*. The effective weights collapse to market shares,* $\tilde{\omega}_j^\ell = \omega_j$*, and the credit multiplier reduces to* $[(1+\eta_B)\mathbf{I} - \mathbf{S}]^{-1}$*. If banks are ex ante identical, the credit distribution is irrelevant for aggregation and the model collapses to a representative-intermediary New Keynesian system augmented by a scalar financial capacity state.*

When $\mathcal{A} = \mathbf{I}$, a bank's fundamental value affects only its own market value. There is no distinction between the bank that holds credit and the bank whose value matters for the rest of the system. With identical banks, reallocating a fixed amount of credit across institutions does not change aggregate financial capacity. The cross-section can be collapsed without losing the macroeconomic mechanism.

When $\mathcal{A} \neq \mathbf{I}$, that collapse is no longer valid. A credit shift toward a bank with a high market share need not have the same macroeconomic effect as a shift toward a bank with a high effective weight. The network contribution is therefore composition dependence: the aggregate effect of monetary policy depends on the identity of the banks that expand, because the network determines how their lending maps into system-wide financial capacity.

This is the sense in which the effective network weight is the central object of the paper. It connects micro-level network positions to the aggregate IS curve, the marginal-cost channel in the NKPC, the output–inflation composition of policy, and the relative amplification of QE and rate policy. The paper's mechanism is not that networks always amplify every shock. It is that networks decide which credit expansions matter most for aggregate outcomes.

5 Ending Remarks

This paper studies how financial networks change the composition of monetary transmission. In a representative-intermediary model, only the aggregate quantity of credit matters. Once banks are connected through balance-sheet linkages, the identity of the banks supplying marginal credit also matters. Monetary policy then affects the economy not only by changing total credit, but by reallocating credit across banks whose network positions imply different aggregate consequences.

The central object is the effective network weight. It links bank-level lending responses to aggregate financial capacity and separates two questions that are identical in a representative-intermediary model: which banks expand after policy easing, and whose

expansion matters most for output and inflation. This distinction generates a financial reallocation channel, helps explain why balance-sheet policies can be amplified differently from interest-rate policy, and creates financial memory through inherited cross-sectional bond holdings.

Several extensions are natural. Empirically, the model points to the measurement of effective network weights and their relationship with observed monetary-policy responses across banks. Theoretically, allowing market shares, network weights, or interbank exposures to evolve away from the steady state would turn inherited financial wedges into richer state dependence. These extensions would sharpen the link between the financial architecture of the banking system and the changing composition of monetary transmission.

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A Proofs for Section 2

A.1 Proof of Proposition 1 (Value Dissipation)

Proposition (Value Dissipation, Formal Statement)**.** *For any cross-holdings matrix* $\mathbf{C} \in \mathbb{R}_+^{n\times n}$ *satisfying (i)* $C_{ii} = 0$ *for all* i*, (ii)* $\hat{C}_{ii} \equiv 1 - \sum_j C_{ji} \in (0, 1]$ *for all* i*, (iii)* $\rho(\mathbf{C}) < 1$*, and any* $\tau \in [0, 1)$*, the dependency matrix* $\mathbf{A}(\tau) = \hat{\mathbf{C}}\,[\mathbf{I} - (1-\tau)\mathbf{C}]^{-1}$ *satisfies*

$$\sum_{i\in\mathcal{N}} A_{ij}(\tau) = 1 - \tau \sum_{i\in\mathcal{N}} \left[\mathbf{C}\,(\mathbf{I} - (1-\tau)\mathbf{C})^{-1}\right]_{ij} \leq 1 \quad \textit{for all } j \in \mathcal{N},$$

with equality if and only if $\tau = 0$ *or* $\mathbf{C} = \mathbf{0}$*. Equivalently, writing* $\mathbf{M}(\tau) \equiv [\mathbf{I} - (1-\tau)\mathbf{C}]^{-1}$*:*

$$\mathbf{1}'\mathbf{A}(\tau) = \mathbf{1}' - \tau\,\mathbf{1}'\mathbf{C}\,\mathbf{M}(\tau).$$

Proof. Write $\mathbf{M} = \mathbf{M}(\tau)$ for brevity. We compute $\mathbf{1}'\mathbf{A}(\tau) = \mathbf{1}'\hat{\mathbf{C}}\,\mathbf{M}$.

The i-th element of the row vector $\mathbf{1}'\hat{\mathbf{C}}$ is $\hat{C}_{ii} = 1 - \sum_j C_{ji}$, so in vector form:

$$\mathbf{1}'\hat{\mathbf{C}} = \mathbf{1}' - \mathbf{1}'\mathbf{C} = \mathbf{1}'(\mathbf{I} - \mathbf{C}).$$

Now decompose $\mathbf{I} - \mathbf{C}$ using the identity $\mathbf{I} - \mathbf{C} = [\mathbf{I} - (1-\tau)\mathbf{C}] - \tau\mathbf{C} = \mathbf{M}^{-1} - \tau\mathbf{C}$:

$$\mathbf{1}'\hat{\mathbf{C}} = \mathbf{1}'(\mathbf{M}^{-1} - \tau\mathbf{C}) = \mathbf{1}'\mathbf{M}^{-1} - \tau\,\mathbf{1}'\mathbf{C}.$$

Multiplying on the right by $\mathbf{M}$:

$$\mathbf{1}'\mathbf{A}(\tau) = \mathbf{1}'\hat{\mathbf{C}}\,\mathbf{M} = \mathbf{1}' - \tau\,\mathbf{1}'\mathbf{C}\,\mathbf{M}.$$

Sign of the dissipation term. Since $\mathbf{C} \geq \mathbf{0}$ elementwise and $\mathbf{M} = \sum_{k=0}^{\infty}[(1-\tau)\mathbf{C}]^k \geq \mathbf{I}$ elementwise, the product $\mathbf{CM} \geq \mathbf{C} \geq \mathbf{0}$. For any j such that $\sum_i C_{ij} > 0$ (i.e., bank j is held by at least one other bank), we have $\sum_i [\mathbf{CM}]_{ij} > 0$, so $\tau \sum_i [\mathbf{CM}]_{ij} > 0$ when $\tau > 0$. Hence $\sum_i A_{ij}(\tau) < 1$ strictly.

Equality conditions. When $\tau = 0$, the dissipation term vanishes identically. When $\mathbf{C} = \mathbf{0}$, $\mathbf{M} = \mathbf{I}$ and $\mathbf{CM} = \mathbf{0}$, so the dissipation term again vanishes. In all other cases with $\tau > 0$ and $\mathbf{C} \neq \mathbf{0}$, the inequality is strict. □

Remark 4. *The dissipation result applies to the dependency matrix* $\mathbf{A}(\tau) = \hat{\mathbf{C}}[\mathbf{I} - (1-\tau)\mathbf{C}]^{-1}$*, where* $\hat{\mathbf{C}}$ *is the outside ownership matrix defined in Section 2.1.1. As emphasized in Section 2.1.3, this result depends only on the general properties of* $\mathbf{A}$ *(non-negativity, column sums bounded by unity) and not on the specific cross-holdings structure used to derive it. Any micro-foundation that generates a dependency matrix with column sums strictly less than one when frictions are present will produce the same qualitative*

conclusion.

A.2 Proof of Proposition 2 (Value Feedback Channel)

Proposition (Value Feedback Channel, Formal Statement)**.** *Under Assumptions 1–2, the semi-elasticity of bank i's credit supply with respect to the aggregate credit spread Δ_t is:*

$$\mathcal{Z}_{i,t} \equiv \frac{\partial \ln B^*_{i,t}}{\partial \Delta_t} = \frac{\phi_{i,t}\, A_{ii}}{1 - \phi_{i,t}\, A_{ii}\, \Delta_t}.$$

This semi-elasticity is (i) strictly positive, (ii) strictly increasing in the dependency index A_{ii}, and (iii) strictly convex in both A_{ii} and Δ_t.

Proof. Taking the natural logarithm of the optimal credit supply (14):

$$\ln B^*_{i,t} = \ln\big[\phi_{i,t}(A_{ii}\, \bar{x}_{i,t} + \bar{v}_{i,t})\big] - \ln Q_t - \ln\big(1 - \phi_{i,t}\, A_{ii}\, \Delta_t\big).$$

Differentiating with respect to Δ_t (treating $\bar{x}_{i,t}$, $\bar{v}_{i,t}$, Q_t, and $\phi_{i,t}$ as exogenous in partial equilibrium):

$$\mathcal{Z}_{i,t} = \frac{\partial \ln B^*_{i,t}}{\partial \Delta_t} = \frac{\phi_{i,t}\, A_{ii}}{1 - \phi_{i,t}\, A_{ii}\, \Delta_t}.$$

Part (i): Positivity. The denominator is positive by Assumption 2 and the numerator is positive since $\phi_{i,t} > 0$ and $A_{ii} > 0$.

Part (ii): Monotonicity in A_{ii}. Differentiating with respect to A_{ii}:

$$\frac{\partial \mathcal{Z}_{i,t}}{\partial A_{ii}} = \frac{\phi_{i,t}}{(1 - \phi_{i,t}\, A_{ii}\, \Delta_t)^2} > 0.$$

Part (iii): Convexity. Differentiating the semi-elasticity a second time:

$$\frac{\partial^2 \mathcal{Z}_{i,t}}{\partial A_{ii}^2} = \frac{2\, \phi_{i,t}^2\, \Delta_t}{(1 - \phi_{i,t}\, A_{ii}\, \Delta_t)^3} > 0,$$

and similarly for Δ_t:

$$\frac{\partial^2 \mathcal{Z}_{i,t}}{\partial \Delta_t^2} = \frac{2\, \phi_{i,t}^2\, A_{ii}^2}{(1 - \phi_{i,t}\, A_{ii}\, \Delta_t)^3} > 0.$$

□

A.3 Proof of Proposition 3 (Policy-Induced Credit Reallocation)

Proposition (Policy-Induced Credit Reallocation, Formal Statement)**.** *Under Assumptions 1–2, the sensitivity of market concentration to an aggregate credit spread shock is:*

$$dHHI_t = 2 \cdot \mathrm{Cov}_{\theta,t}(\theta_{i,t},\, \mathcal{Z}_{i,t}) \cdot d\Delta_t.$$

Since $\mathcal{Z}_{i,t}$ is strictly increasing in A_{ii} (Proposition 2), concentration rises in response to accommodative policy ($d\Delta_t > 0$) if and only if $\mathrm{Cov}_{\theta,t}(\theta_{i,t}, A_{ii}) > 0$.

Proof. The change in bank i's market share satisfies:

$$d\ln\theta_{i,t} = d\ln B_{i,t} - d\ln B_t = \left(\mathcal{Z}_{i,t} - \bar{\mathcal{Z}}_t\right) d\Delta_t,$$

where $\bar{\mathcal{Z}}_t \equiv \mathbb{E}_{\theta,t}[\mathcal{Z}_{j,t}]$ is the share-weighted average semi-elasticity. The second equality uses the fact that the growth rate of aggregate credit is the share-weighted average of individual growth rates:

$$d\ln B_t = \sum_{j\in\mathcal{N}} \frac{B_{j,t}}{B_t}\, d\ln B_{j,t} = \sum_{j\in\mathcal{N}} \theta_{j,t}\, \mathcal{Z}_{j,t}\, d\Delta_t = \bar{\mathcal{Z}}_t\, d\Delta_t.$$

Differentiating $HHI_t = \sum_i \theta_{i,t}^2$:

$$\begin{aligned} dHHI_t &= 2\sum_i \theta_{i,t}\, d\theta_{i,t} = 2\sum_i \theta_{i,t}^2\left(\mathcal{Z}_{i,t} - \bar{\mathcal{Z}}_t\right) d\Delta_t \\ &= 2\Big[\mathbb{E}_{\theta,t}[\theta_{i,t}\,\mathcal{Z}_{i,t}] - \mathbb{E}_{\theta,t}[\theta_{i,t}]\,\mathbb{E}_{\theta,t}[\mathcal{Z}_{i,t}]\Big]\, d\Delta_t \\ &= 2\,\mathrm{Cov}_{\theta,t}(\theta_{i,t},\, \mathcal{Z}_{i,t})\, d\Delta_t. \end{aligned}$$

The final claim follows from the monotonicity of $\mathcal{Z}_{i,t}$ in A_{ii} (Proposition 2): a positive covariance between $\theta_{i,t}$ and $\mathcal{Z}_{i,t}$ obtains if and only if $\mathrm{Cov}_{\theta,t}(\theta_{i,t}, A_{ii}) > 0$, since $\mathcal{Z}_{i,t}$ is a strictly increasing transformation of A_{ii} (holding $\phi_{i,t}$ constant). □

Corollary (Market Share Reallocation, Formal Statement)**.** *In response to a marginal increase in the credit spread $d\Delta_t > 0$, bank i's market share grows ($d\theta_{i,t} > 0$) if and only if its semi-elasticity exceeds the share-weighted average:*

$$d\ln\theta_{i,t} = \left(\mathcal{Z}_{i,t} - \bar{\mathcal{Z}}_t\right) d\Delta_t.$$

Since $\mathcal{Z}_{i,t}$ is increasing in A_{ii}, capital flows toward banks with higher dependency indices.

Proof. By definition, $\ln\theta_{i,t} = \ln B_{i,t} - \ln B_t$. Differentiating with respect to the aggregate shock:

$$d\ln\theta_{i,t} = d\ln B_{i,t} - d\ln B_t.$$

Substituting $d\ln B_{i,t} = \mathcal{Z}_{i,t}\, d\Delta_t$ and $d\ln B_t = \bar{\mathcal{Z}}_t\, d\Delta_t$ yields the result. The condition for reallocation follows directly from the sign of $(\mathcal{Z}_{i,t} - \bar{\mathcal{Z}}_t)$, which by Proposition 2 is positive if and only if A_{ii} exceeds the share-weighted average dependency index (holding $\phi_{i,t}$ constant). □

A.4 Proof of Proposition 5 (Symmetric vs. Asymmetric Networks)

Proposition (Symmetric vs. Asymmetric Networks, Formal Statement)**.** *Consider the class of networks parameterized by a uniform cross-holding intensity $\delta \in [0, 1)$ and a topology matrix $\mathbf{G}$ with $G_{ii} = 0$ and $\sum_j G_{ij} = 1$ for all i, so that $\mathbf{C} = \delta\mathbf{G}$.*

(a) Symmetric networks. *If $\mathbf{G}$ is vertex-transitive, then $A_{ii} = A_{jj}$ for all i, j, the semi-elasticities $\mathcal{Z}_{i,t}$ are equalized across banks conditional on common $\phi_{i,t}$, and $dHHI_t = 0$.*

(b) Asymmetric networks. *If banks occupy non-equivalent network positions, dependency indices generically differ across banks. The direction of reallocation is governed by $\mathrm{Cov}_{\theta,t}(\theta_{i,t}, A_{ii})$.*

Proof. If $\mathbf{G}$ is vertex-transitive, for any banks i and j there is a permutation matrix $\mathbf{P}_{ij}$ such that $\mathbf{P}_{ij}\mathbf{G}\mathbf{P}'_{ij} = \mathbf{G}$ and $\mathbf{P}_{ij}e_i = e_j$. Hence the same permutation leaves $\mathbf{C} = \delta\mathbf{G}$, outside ownership, and $[\mathbf{I} - (1-\tau)\mathbf{C}]^{-1}$ invariant. The diagonal entries of $\mathbf{A} = \hat{\mathbf{C}}[\mathbf{I} - (1-\tau)\mathbf{C}]^{-1}$ are therefore identical. With common $\phi_{i,t}$, the semi-elasticities are identical and reallocation is zero. If banks are not in equivalent network positions, the outside ownership shares or the diagonal entries of the network inverse generically differ, so A_{ii} differs across banks and Proposition 3 gives the reallocation result. □

B Proofs for Section 4

B.1 Proof of Proposition 6 (Financial Reallocation and Output–Inflation Composition)

Proposition (Financial Reallocation and Output–Inflation Composition, Formal Statement)**.** *Consider a monetary shock and define*

$$\gamma_t \equiv \left.\frac{\partial \Theta_t^{net}}{\partial \hat{y}_t}\right|_{monetary\ shock}.$$

The Phillips Curve can be written, for the component of inflation that co-moves with output under this policy experiment, as

$$\hat{\pi}_t = \beta\mathbb{E}_t\hat{\pi}_{t+1} + \tilde{\kappa}_t\hat{y}_t + \tilde{u}_t, \qquad \tilde{\kappa}_t = \kappa_y - \Lambda_C\bar{b}^{FI}\gamma_t.$$

If $\gamma_t > 0$, the co-movement of output with network financial capacity creates a marginal-cost offset to the inflationary effect of higher output. Comparing a network economy and a no-network economy under the same policy experiment, monetary transmission shifts

further toward real activity and away from inflation in the network economy if and only if

$$\gamma_t^{\text{net}} > \gamma_t^{\text{no net}}.$$

This condition is more likely when banks with large policy responses also carry large effective network weights.

Proof. Start from the augmented Phillips Curve in equation (33):

$$\hat{\pi}_t = \beta \mathbb{E}_t \hat{\pi}_{t+1} + \kappa_y \hat{y}_t - \Lambda_C \big(\bar{b}^{FI} \Theta_t^{\text{net}} + \bar{b}^{CB} \widehat{qe}_t\big) - \psi_a \hat{a}_t.$$

For the monetary experiment under consideration, decompose network financial capacity into the component that co-moves with output and the component orthogonal to output:

$$\Theta_t^{\text{net}} = \gamma_t \hat{y}_t + \Theta_t^{\perp}, \qquad \left.\frac{\partial \Theta_t^{\text{net}}}{\partial \hat{y}_t}\right|_{\text{monetary shock}} = \gamma_t.$$

Substituting this decomposition into the Phillips Curve and collecting terms proportional to $\hat{y}_t$ gives

$$\hat{\pi}_t = \beta \mathbb{E}_t \hat{\pi}_{t+1} + \big(\kappa_y - \Lambda_C \bar{b}^{FI} \gamma_t\big) \hat{y}_t + \tilde{u}_t,$$

where

$$\tilde{u}_t = -\Lambda_C \bar{b}^{FI} \Theta_t^{\perp} - \Lambda_C \bar{b}^{CB} \widehat{qe}_t - \psi_a \hat{a}_t$$

collects terms not proportional to $\hat{y}_t$. This proves the stated representation with $\tilde{\kappa}_t = \kappa_y - \Lambda_C \bar{b}^{FI} \gamma_t$.

Because $\Lambda_C > 0$ and $\bar{b}^{FI} > 0$, a positive γ_t lowers the inflation response associated with a given output movement relative to the case in which financial capacity is held fixed. Equivalently, the financial-capacity movement produces the supply-side credit offset described in the main text.

Now compare the network and no-network economies. Holding fixed the mechanical preference and pricing parameters $(\kappa_y, \Lambda_C, \bar{b}^{FI})$, the difference in the effective output coefficient is

$$\tilde{\kappa}_t^{\text{net}} - \tilde{\kappa}_t^{\text{no net}} = -\Lambda_C \bar{b}^{FI} \big(\gamma_t^{\text{net}} - \gamma_t^{\text{no net}}\big).$$

Hence

$$\tilde{\kappa}_t^{\text{net}} < \tilde{\kappa}_t^{\text{no net}} \quad \Longleftrightarrow \quad \gamma_t^{\text{net}} > \gamma_t^{\text{no net}}.$$

Since a smaller output coefficient means that a given output response is accompanied by a smaller inflation response, the condition is exactly the statement that monetary transmission shifts further toward real activity and away from inflation.

It remains to connect the condition to reallocation. From equation (51), the response of Θ_t^{net} to spreads, bond prices, and bank-level shocks is governed by effective network weights and by the credit multiplier embedded in $\mathbf{\Gamma}'$. Proposition 2 identifies which banks

respond most to easing; the aggregation equation identifies how much those responses move aggregate financial capacity. Thus γ_t^{net} is larger when high-response banks are also high-effective-weight banks. This proves the alignment statement. □

B.2 Proof of Proposition 7 (Conditional Asymmetric Amplification)

Proposition (Conditional Asymmetric Network Amplification, Formal Statement)**.** *Let the output response of each policy instrument be decomposed into a direct household Euler-equation component and a bank balance-sheet component. Suppose the network multiplies the balance-sheet component of both instruments by a common factor $\rho > 1$, and suppose the interest-rate instrument has a positive direct household component that is invariant to the network. If QE loads only on the balance-sheet component in the comparison, then the output amplification ratio for QE exceeds the output amplification ratio for conventional interest-rate policy.*

Proof. Let $B_r > 0$ denote the no-network balance-sheet component of the output response to interest-rate policy, and let $B_q > 0$ denote the no-network balance-sheet component of the output response to QE. Let $H > 0$ denote the direct household Euler-equation component of interest-rate policy. This component is a household intertemporal-substitution response and is therefore independent of the financial network. Under the maintained assumption, the network multiplies balance-sheet components by the common factor $\rho > 1$:

$$B_r^{\text{net}} = \rho B_r, \qquad B_q^{\text{net}} = \rho B_q.$$

For QE, the output response in this comparison is entirely balance-sheet based, so its amplification ratio is

$$\mathcal{R}_q = \frac{B_q^{\text{net}}}{B_q} = \rho.$$

For interest-rate policy, the output response contains both the direct household bypass and the balance-sheet component. Its amplification ratio is

$$\mathcal{R}_r = \frac{H + \rho B_r}{H + B_r}.$$

Since $H > 0$ and $\rho > 1$,

$$H + \rho B_r < \rho H + \rho B_r = \rho(H + B_r),$$

so

$$\mathcal{R}_r = \frac{H + \rho B_r}{H + B_r} < \rho = \mathcal{R}_q.$$

The inequality is strict because the household bypass is positive. Thus the instrument with the larger balance-sheet share, QE, receives the larger overall network amplification even when the network multiplies the common balance-sheet component by the same factor for both instruments. □

B.3 Proof of Proposition 8 (Financial Memory Through Inherited Bond Holdings)

Proposition (Financial Memory Through Inherited Bond Holdings, Formal Statement)**.** *In the first-order system with fixed valuation-elasticity matrix $\mathcal{A}$ and fixed effective network weights $\tilde{\omega}^{\ell}$, inherited bank bond holdings affect current network financial capacity through the term $\mathbf{\Psi}'\hat{\mathbf{b}}_{t-1}^{FI}$ in equation (51). The cross-section of bank bond holdings evolves according to*

$$\hat{\mathbf{b}}_t^{FI} = \mathbf{P}\hat{\mathbf{b}}_{t-1}^{FI} + \textit{current shocks and prices}, \qquad \mathbf{P} = [(1+\eta_B)\mathbf{I} - \mathcal{A}\mathbf{S}]^{-1}(\mathcal{A}\mathbf{E} + \eta_B\mathbf{I}).$$

Consequently, past monetary policy affects the future financial position from which new shocks are transmitted. The persistence of this inherited financial wedge is increasing in $\rho(\mathbf{P})$.

Proof. From equation (51), current network financial capacity can be written as

$$\Theta_t^{\text{net}} = \hat{\phi}_t^{agg} + \mathbf{\Gamma}'\hat{\boldsymbol{\phi}}_t + \mu^*\hat{\Delta}_t + \alpha^*\hat{q}_t + \mathbf{\Psi}'\hat{\mathbf{b}}_{t-1}^{FI} - \eta_B(\hat{b}_t^{FI} - \hat{b}_{t-1}^{FI}),$$

where

$$\mathbf{\Psi}' = \mathbf{\Lambda}^{(1)} + \mathbf{\Gamma}'(\mathcal{A}\mathbf{E} + \eta_B\mathbf{I}).$$

The term $\mathbf{\Psi}'\hat{\mathbf{b}}_{t-1}^{FI}$ is the direct effect of inherited cross-sectional bond holdings on current financial capacity. It is cross-sectional: for a fixed aggregate $\hat{b}_{t-1}^{FI}$, different allocations of inherited bond holdings across banks can generate different values of this term whenever the components of $\mathbf{\Psi}$ differ across banks.

The law of motion for credit in equation (36) can be written as

$$\hat{\mathbf{b}}_t^{FI} = [(1+\eta_B)\mathbf{I} - \mathcal{A}\mathbf{S}]^{-1}\big[\hat{\boldsymbol{\phi}}_t + \boldsymbol{\alpha}_q\hat{q}_t + \boldsymbol{\alpha}_\Delta\hat{\Delta}_t\big] + \mathbf{P}\hat{\mathbf{b}}_{t-1}^{FI},$$

with $\mathbf{P}$ as stated. Iterating forward from an inherited distribution at date $t-1$ gives

$$\hat{\mathbf{b}}_{t+h}^{FI} = \mathbf{P}^{h+1}\hat{\mathbf{b}}_{t-1}^{FI} + \sum_{m=0}^{h}\mathbf{P}^m\big(\text{current shocks and prices at } t+h-m\big).$$

Thus the effect of an inherited allocation on future credit distributions is governed by powers of $\mathbf{P}$.

By the Gelfand formula,

$$\lim_{h\to\infty} \|\mathbf{P}^h\|^{1/h} = \rho(\mathbf{P})$$

for any matrix norm. Hence a larger spectral radius implies slower decay of inherited cross-sectional positions, conditional on stability $\rho(\mathbf{P}) < 1$. Since inherited positions enter financial capacity through $\mathbf{\Psi}'\hat{\mathbf{b}}^{FI}_{t-1}$, this establishes that the persistence of the inherited financial wedge is increasing in $\rho(\mathbf{P})$.

Finally, because $\mathcal{A}$ and $\tilde{\omega}^{\ell}$ are fixed around the steady state in the first-order approximation, the result is not a change in local structural coefficients. It is a shift in the inherited financial wedge and therefore in the financial position from which future policy shocks are transmitted. □

B.4 Proof of Proposition 9 (Degenerate Network Limit)

Proposition (Degenerate Network Limit, Formal Statement)**.** *When* $\mathbf{C} = \mathbf{0}$*, the level dependency matrix is* $\mathbf{A} = \mathbf{I}$ *and the valuation-elasticity matrix is* $\mathcal{A} = \mathbf{I}$*. The effective network weights collapse to market shares,*

$$\tilde{\omega}^{\ell}_j = \omega_j,$$

and the credit multiplier reduces to the diagonal matrix

$$[(1+\eta_B)\mathbf{I} - \mathbf{S}]^{-1}.$$

If banks are ex ante identical, the cross-sectional credit distribution is irrelevant for aggregation and the model collapses to a representative-intermediary New Keynesian system augmented by a scalar financial-capacity state.

Proof. Dependency and valuation matrices. If $\mathbf{C} = \mathbf{0}$, then outside ownership is complete: $\hat{\mathbf{C}} = \mathbf{I}$. Therefore

$$\mathbf{A} = \hat{\mathbf{C}}[\mathbf{I} - (1-\tau)\mathbf{C}]^{-1} = \mathbf{I}.$$

Since $\mathbf{v}_t = \mathbf{x}_t$ in this limit, $\bar{v}_i = \bar{x}_i$ for all i. The valuation-elasticity matrix satisfies

$$\mathcal{A}_{ij} = \frac{A_{ij}\bar{x}_j}{\bar{v}_i} = \frac{\delta_{ij}\bar{x}_j}{\bar{x}_i} = \delta_{ij},$$

so $\mathcal{A} = \mathbf{I}$.

Effective weights. Using the definition of the effective network weight,

$$\tilde{\omega}^{\ell}_j = \sum_i \omega_i \mathcal{A}_{ij} = \sum_i \omega_i \delta_{ij} = \omega_j.$$

Thus there is no distinction between a bank's market share and its aggregate consequence

weight.

Credit multiplier. Substituting $\mathcal{A} = \mathbf{I}$ into the credit distribution block gives

$$\hat{\mathbf{b}}_t^{FI} = [(1+\eta_B)\mathbf{I} - \mathbf{S}]^{-1}\Big[\hat{\boldsymbol{\phi}}_t + \boldsymbol{\alpha}_q \hat{q}_t + \boldsymbol{\alpha}_\Delta \hat{\Delta}_t + (\mathbf{E} + \eta_B \mathbf{I})\hat{\mathbf{b}}_{t-1}^{FI}\Big].$$

Because $\mathbf{S}$ and $\mathbf{E}$ are diagonal, the multiplier $[(1+\eta_B)\mathbf{I} - \mathbf{S}]^{-1}$ is diagonal. Bank-level lending dynamics are therefore uncoupled: a shock to bank j affects bank j's credit through its own balance sheet, but there is no network propagation to other banks.

Aggregation. With $\tilde{\omega}_j^\ell = \omega_j$, the aggregation coefficients in equation (41) become

$$\bar{\mu} = \sum_j \omega_j s_j, \qquad \bar{\alpha} = \sum_j \omega_j (s_j + e_j), \qquad \Lambda_j^{(0)} = \omega_j s_j, \qquad \Lambda_j^{(1)} = \omega_j e_j.$$

If banks are ex ante identical, then $s_j = s$, $e_j = e$, and $\phi_{j,t} = \phi_t$ for all j. Since $\sum_j \omega_j = 1$,

$$\bar{\mu} = s, \qquad \bar{\alpha} = s + e, \qquad \boldsymbol{\Lambda}^{(0)}\hat{\mathbf{b}}_t^{FI} = s\sum_j \omega_j \hat{b}_{j,t} = s\hat{b}_t^{FI},$$

and similarly

$$\boldsymbol{\Lambda}^{(1)}\hat{\mathbf{b}}_{t-1}^{FI} = e\hat{b}_{t-1}^{FI}.$$

Thus the network aggregation equation depends on the banking sector only through the scalar aggregate $\hat{b}_t^{FI}$ and the scalar financial capacity $\Theta_t^{\text{net}} = \hat{q}_t + \hat{b}_t^{FI}$.

The cross-sectional distribution is therefore irrelevant in the degenerate, ex ante identical limit. Substituting the scalar financial capacity state into the IS curve and the Phillips Curve yields the representative-intermediary New Keynesian system augmented by one financial-capacity state, as claimed. □

C Model Appendix: Settings, First-Order Conditions, and Reduction

This appendix states the complete model environment, first-order conditions, market-clearing conditions, and the algebraic reduction to the four-equation system used in the paper. Throughout, hatted variables denote log deviations from the non-stochastic steady state, $\hat{x}_t \equiv \log(x_t/\bar{x})$.

The level dependency matrix and its log-linear counterpart are distinct:

$$\mathbf{A}(\tau) = \hat{\mathbf{C}}[\mathbf{I} - (1-\tau)\mathbf{C}]^{-1}, \qquad \mathcal{A}_{ij} \equiv \frac{A_{ij}\bar{x}_j}{\bar{v}_i}. \tag{55}$$

The matrix $\mathbf{A}$ maps fundamental value levels into outside market-value levels, while $\mathcal{A}$ maps log deviations:

$$\mathbf{v}_t = \mathbf{A}\mathbf{x}_t, \qquad \hat{v}_{i,t} = \sum_j \mathcal{A}_{ij}\hat{x}_{j,t}. \tag{56}$$

C.1 Timing, Households, and Asset Pricing

The economy is in discrete time. At the beginning of period t, the previous generation of bankers exits and distributes dividends. New bankers enter and receive equity transfers. Households choose consumption, labor, saving, and borrowing. Banks choose long-bond and reserve portfolios subject to leverage. Firms produce and Calvo-adjusting retailers reset prices. The central bank sets the short rate and its bond portfolio according to policy rules. Returns are realized between t and $t+1$.

The long bond bought at t pays one unit at $t+1$, κ at $t+2$, and so on. Its gross nominal holding-period return is

$$R^b_{t+1} = \frac{1+\kappa Q_{t+1}}{Q_t}. \tag{57}$$

Patient household. The patient household consumes $C_{P,t}$, supplies labor L_t, saves through nominal deposits, owns production-firm equity, and owns outside equity claims on financial intermediaries. Preferences are

$$\mathbb{E}_t \sum_{j=0}^{\infty} \beta^j \left[\frac{C_{P,t+j}^{1-\sigma} - 1}{1-\sigma} - \psi \frac{L_{t+j}^{1+\chi}}{1+\chi} \right]. \tag{58}$$

The nominal budget constraint is

$$\begin{aligned} P_t C_{P,t} + \sum_i S_{i,t} + \sum_i P_t X^{FI}_{i,t} + P_t X^b_t \leq W_t L_t + \sum_i R^s_{t-1} S_{i,t-1} \\ + P_t D^F_t + P_t D^{P,FI}_t + P_t T_t. \end{aligned} \tag{59}$$

Let Λ_t^P denote the multiplier on the nominal budget constraint. The consumption and labor first-order conditions are

$$C_{P,t}^{-\sigma} = \Lambda_t^P P_t, \tag{60}$$

$$\psi L_t^\chi = \Lambda_t^P W_t. \tag{61}$$

Combining them and defining $w_t = W_t/P_t$ gives

$$\psi L_t^\chi = w_t C_{P,t}^{-\sigma}. \tag{62}$$

The deposit FOC is

$$\Lambda_t^P = \beta \mathbb{E}_t[\Lambda_{t+1}^P R_t^s]. \tag{63}$$

Using the consumption FOC,

$$1 = \mathbb{E}_t \left[\beta \left(\frac{C_{P,t+1}}{C_{P,t}} \right)^{-\sigma} \frac{R_t^s}{\Pi_{t+1}} \right]. \tag{64}$$

The nominal stochastic discount factor is

$$M_{t,t+1}^N = \beta \left(\frac{C_{P,t+1}}{C_{P,t}} \right)^{-\sigma} \frac{1}{\Pi_{t+1}}. \tag{65}$$

Impatient household. The impatient household consumes $C_{B,t}$, issues long-term bonds, and has discount factor $\beta_B < \beta$. Preferences are

$$\mathbb{E}_t \sum_{j=0}^{\infty} \beta_B^j \frac{C_{B,t+j}^{1-\sigma} - 1}{1-\sigma}. \tag{66}$$

The nominal budget constraint is

$$P_t C_{B,t} + B_{t-1}^C \leq Q_t (B_t^C - \kappa B_{t-1}^C) + P_t X_t^b. \tag{67}$$

The transfer rule is

$$P_t X_t^b = (1 + \kappa Q_t) B_{t-1}^C. \tag{68}$$

Substituting (68) into (67) gives

$$P_t C_{B,t} = Q_t B_t^C. \tag{69}$$

The borrower optimality condition with a debt-elastic closing device is

$$1 = \beta_B \left(\frac{C_{B,t+1}}{C_{B,t}} \right)^{-\sigma} \frac{R_{t+1}^b}{\Pi_{t+1}} - \psi_b \left(\frac{b_{C,t}}{\bar{b}_C} - 1 \right). \tag{70}$$

C.2 Financial Intermediaries

There are n banks indexed by $i \in \mathcal{N} = \{1, \ldots, n\}$. The cross-holdings matrix $\mathbf{C}$ has elements $C_{ij} \in [0, 1)$, where C_{ij} is the fraction of bank j's equity held by bank i. The restrictions are

$$C_{ii} = 0, \qquad \hat{C}_{ii} \equiv 1 - \sum_j C_{ji} \in (0, 1], \qquad \rho(\mathbf{C}) < 1. \tag{71}$$

Let $\hat{\mathbf{C}} = \mathrm{diag}(\hat{C}_{11}, \ldots, \hat{C}_{nn})$. The interbank dividend-chain friction is $\tau \in [0, 1)$: each ownership link transmits only $(1 - \tau)$ of the value it receives.

Network valuation. Fundamental value is

$$x_{i,t} = \mathbb{E}_t[M^N_{t,t+1} D^{FI}_{i,t+1}] = \Delta_t Q_t B_{i,t} + \bar{x}_{i,t}, \tag{72}$$

where

$$\Delta_t \equiv \mathbb{E}_t[M^N_{t,t+1}(R^b_{t+1} - R^s_t)]. \tag{73}$$

Total equity values satisfy

$$\mathbf{V}_t = \mathbf{x}_t + (1 - \tau)\mathbf{C}\mathbf{V}_t, \qquad \mathbf{V}_t = [\mathbf{I} - (1 - \tau)\mathbf{C}]^{-1}\mathbf{x}_t. \tag{74}$$

Outside market value is

$$\mathbf{v}_t = \hat{\mathbf{C}}\mathbf{V}_t = \mathbf{A}(\tau)\mathbf{x}_t, \qquad \mathbf{A}(\tau) \equiv \hat{\mathbf{C}}[\mathbf{I} - (1 - \tau)\mathbf{C}]^{-1}. \tag{75}$$

Its log-linearization is

$$\hat{v}_{i,t} = \sum_j \mathcal{A}_{ij} \hat{x}_{j,t}, \qquad \mathcal{A}_{ij} \equiv \frac{A_{ij}\bar{x}_j}{\bar{v}_i}. \tag{76}$$

Balance sheet, transfers, and dividends. The bank balance sheet is

$$Q_t B_{i,t} + RE_{i,t} = S_{i,t} + P_t X^{FI}_{i,t}. \tag{77}$$

Equity transfers are

$$P_t X^{FI}_{i,t} = P_t \bar{X}^{FI}_i + \kappa Q_t B_{i,t-1}. \tag{78}$$

The fundamental dividend paid by the exiting generation is

$$\begin{aligned} P_t D^{FI}_{i,t} = {} & (R^b_t - R^s_{t-1}) Q_{t-1} B_{i,t-1} + (R^{re}_{t-1} - R^s_{t-1}) RE_{i,t-1} \\ & + R^s_{t-1} P_t X^{FI}_{i,t-1}. \end{aligned} \tag{79}$$

Network dividend distribution is

$$\tilde{\mathbf{D}}_t^{FI} = [\mathbf{I} - (1-\tau)\mathbf{C}]^{-1}\mathbf{D}_t^{FI}. \tag{80}$$

The patient household receives

$$D_t^{P,FI} = \mathbf{1}'\mathbf{A}\mathbf{D}_t^{FI}. \tag{81}$$

Bank problem and KKT conditions. The leverage constraint is

$$Q_t B_{i,t} \leq \phi_{i,t} v_{i,t} - \eta_B \bar{Q}(B_{i,t} - B_{i,t-1}). \tag{82}$$

Under fixed spillovers,

$$v_{i,t} = A_{ii}(\Delta_t Q_t B_{i,t} + \bar{x}_{i,t}) + \bar{v}_{i,t}, \qquad \bar{v}_{i,t} \equiv \sum_{j \neq i} A_{ij} x_{j,t}. \tag{83}$$

Bank i chooses $B_{i,t} \geq 0$ to maximize $v_{i,t}$ subject to (82). Define

$$g_{i,t} \equiv \phi_{i,t} v_{i,t} - \eta_B \bar{Q}(B_{i,t} - B_{i,t-1}) - Q_t B_{i,t} \geq 0. \tag{84}$$

The Lagrangian is

$$\mathcal{L}_{i,t}^{FI} = v_{i,t} + \Omega_{i,t} g_{i,t} + \nu_{i,t} B_{i,t}. \tag{85}$$

The KKT conditions are

$$\Omega_{i,t} \geq 0, \qquad \nu_{i,t} \geq 0, \tag{86}$$

$$g_{i,t} \geq 0, \qquad B_{i,t} \geq 0, \tag{87}$$

$$\Omega_{i,t} g_{i,t} = 0, \qquad \nu_{i,t} B_{i,t} = 0, \tag{88}$$

$$A_{ii}\Delta_t Q_t + \Omega_{i,t}\left(\phi_{i,t} A_{ii} \Delta_t Q_t - \eta_B \bar{Q} - Q_t\right) + \nu_{i,t} = 0. \tag{89}$$

For an interior bond position, $\nu_{i,t} = 0$, hence

$$\Omega_{i,t}\left[Q_t + \eta_B \bar{Q} - \phi_{i,t} A_{ii} \Delta_t Q_t\right] = A_{ii} \Delta_t Q_t. \tag{90}$$

When $\Delta_t > 0$, the objective is increasing in $B_{i,t}$, so the leverage constraint binds. Solving the binding constraint gives

$$B_{i,t}^* = \frac{\phi_{i,t}(A_{ii}\bar{x}_{i,t} + \bar{v}_{i,t}) + \eta_B \bar{Q} B_{i,t-1}}{Q_t(1 - \phi_{i,t} A_{ii} \Delta_t) + \eta_B \bar{Q}}. \tag{91}$$

The bank-side stability restrictions are

$$1 - \phi_{i,t} A_{ii} \Delta_t > 0, \qquad \rho((1+\eta_B)^{-1}\mathcal{A}\mathbf{S}) < 1. \tag{92}$$

C.3 Firms, Policy, Market Clearing, and Equilibrium

The final good firm aggregates differentiated retail varieties:

$$Y_t = \left(\int_0^1 Y_t(f)^{\frac{\epsilon-1}{\epsilon}} df \right)^{\frac{\epsilon}{\epsilon-1}}, \qquad \epsilon > 1. \tag{93}$$

Profit maximization gives demand and the price index:

$$Y_t(f) = \left(\frac{P_t(f)}{P_t} \right)^{-\epsilon} Y_t, \qquad P_t = \left(\int_0^1 P_t(f)^{1-\epsilon} df \right)^{\frac{1}{1-\epsilon}}. \tag{94}$$

Retailers buy wholesale output at nominal marginal cost $P_{m,t}$. Each period, fraction $1-\theta_p$ resets its price. An adjusting retailer solves

$$\max_{P_t^*} \mathbb{E}_t \sum_{k=0}^{\infty} \theta_p^k M_{t,t+k} \left[P_t^* Y_{t+k|t} - P_{m,t+k} Y_{t+k|t} \right], \tag{95}$$

subject to

$$Y_{t+k|t} = \left(\frac{P_t^*}{P_{t+k}} \right)^{-\epsilon} Y_{t+k}. \tag{96}$$

The reset-price FOC is

$$\mathbb{E}_t \sum_{k=0}^{\infty} \theta_p^k M_{t,t+k} Y_{t+k|t} \left[P_t^* - \frac{\epsilon}{\epsilon-1} P_{m,t+k} \right] = 0. \tag{97}$$

Equivalently,

$$p_t^* = \frac{\epsilon}{\epsilon-1} \frac{F_t}{K_t}, \tag{98}$$

$$F_t = Y_t p_{m,t} + \beta \theta_p \mathbb{E}_t [\Pi_{t+1}^{\epsilon} F_{t+1}], \tag{99}$$

$$K_t = Y_t + \beta \theta_p \mathbb{E}_t [\Pi_{t+1}^{\epsilon-1} K_{t+1}]. \tag{100}$$

The price-index recursion and price-dispersion law are

$$1 = \theta_p \Pi_t^{\epsilon-1} + (1-\theta_p)(p_t^*)^{1-\epsilon}, \tag{101}$$

$$\nu_t^p = \theta_p \Pi_t^{\epsilon} \nu_{t-1}^p + (1-\theta_p)(p_t^*)^{-\epsilon}. \tag{102}$$

Log-linearization around zero inflation gives

$$\hat{\pi}_t = \beta \mathbb{E}_t \hat{\pi}_{t+1} + \lambda_p \widehat{mc}_t, \qquad \lambda_p = \frac{(1-\theta_p)(1-\theta_p\beta)}{\theta_p}. \tag{103}$$

Wholesale production is

$$Y_{m,t} = A_t L_t, \qquad \ln A_t = \rho_A \ln A_{t-1} + \varepsilon_{A,t}. \tag{104}$$

Cost minimization gives

$$w_t = p_{m,t} A_t. \tag{105}$$

Aggregate production-firm dividends are

$$D_t^F = Y_t - w_t L_t. \tag{106}$$

Monetary policy is rule-based:

$$\frac{R_t^s}{R^s} = \left(\frac{R_{t-1}^s}{R^s}\right)^{\rho_r} \left[\left(\frac{\Pi_t}{\Pi^*}\right)^{\phi_\pi} \left(\frac{Y_t}{Y_t^*}\right)^{\phi_y}\right]^{1-\rho_r} e^{\varepsilon_{r,t}}, \tag{107}$$

$$\frac{QE_t}{QE} = \left(\frac{QE_{t-1}}{QE}\right)^{\rho_q} \left[\left(\frac{\Pi_t}{\Pi^*}\right)^{-\lambda_\pi} \left(\frac{Y_t}{Y_t^*}\right)^{-\lambda_y}\right]^{1-\rho_q} e^{\varepsilon_{q,t}}. \tag{108}$$

The central bank balance sheet, real QE definition, interest on reserves, and surplus transfer are

$$Q_t B_t^{cb} = RE_t, \tag{109}$$

$$QE_t \equiv \frac{Q_t B_t^{cb}}{P_t} = q_t b_t^{cb}, \tag{110}$$

$$R_t^{re} = R_t^s, \tag{111}$$

$$P_t T_t = R_t^b Q_{t-1} B_{t-1}^{cb} - R_{t-1}^{re} RE_{t-1}. \tag{112}$$

Market clearing requires

$$B_t^C = \sum_i B_{i,t} + B_t^{cb} = B_t^{FI} + B_t^{cb}, \tag{113}$$

$$RE_t = \sum_i RE_{i,t} = RE_t^{FI}, \tag{114}$$

$$S_t = \sum_i S_{i,t} = S_t^{FI}, \tag{115}$$

$$L_t^s = L_t^d = L_t, \tag{116}$$

$$Y_t = C_{P,t} + C_{B,t}. \tag{117}$$

Using (69), the aggregate resource constraint is

$$Y_t = C_{P,t} + q_t b_t^C. \tag{118}$$

Aggregate production with price dispersion is

$$Y_t \nu_t^p = A_t L_t. \tag{119}$$

An equilibrium is a sequence of allocations, prices, bank values, and policy variables such that households optimize, banks maximize market value subject to leverage, firms maximize profits, the central bank follows the policy rules, values and dividends satisfy the network equations, and all markets clear.

C.4 Step-by-Step Reduction to the Four-Equation System

C.4.1 Aggregate demand

Goods clearing log-linearizes to

$$\hat{y}_t = (1 - z_C)\hat{c}_{P,t} + z_C \hat{c}_{B,t}, \qquad z_C \equiv \frac{\bar{C}_B}{\bar{Y}}. \tag{120}$$

The patient Euler equation log-linearizes to

$$\hat{c}_{P,t} = \mathbb{E}_t \hat{c}_{P,t+1} - \frac{1}{\sigma}\left(\hat{r}_t^s - \mathbb{E}_t \hat{\pi}_{t+1}\right). \tag{121}$$

In real terms, (69) gives

$$C_{B,t} = q_t B_t^C, \qquad \hat{c}_{B,t} = \hat{q}_t + \hat{b}_t^C. \tag{122}$$

Bond clearing log-linearizes to

$$\hat{b}_t^C = \bar{b}^{FI}\hat{b}_t^{FI} + \bar{b}^{CB}\hat{b}_t^{CB}, \qquad \bar{b}^{FI} + \bar{b}^{CB} = 1. \tag{123}$$

Define

$$\Theta_t^{\text{net}} \equiv \hat{q}_t + \hat{b}_t^{FI}, \qquad \widehat{qe}_t \equiv \hat{q}_t + \hat{b}_t^{CB}. \tag{124}$$

Then

$$\begin{aligned}
\hat{c}_{B,t} &= \hat{q}_t + \bar{b}^{FI}\hat{b}_t^{FI} + \bar{b}^{CB}\hat{b}_t^{CB} \\
&= \bar{b}^{FI}(\hat{q}_t + \hat{b}_t^{FI}) + \bar{b}^{CB}(\hat{q}_t + \hat{b}_t^{CB}) \\
&= \bar{b}^{FI}\Theta_t^{\text{net}} + \bar{b}^{CB}\widehat{qe}_t.
\end{aligned} \tag{125}$$

Write goods clearing at t and $t+1$:

$$\hat{y}_t = (1-z_C)\hat{c}_{P,t} + z_C\left[\bar{b}^{FI}\Theta_t^{\text{net}} + \bar{b}^{CB}\widehat{qe}_t\right], \tag{126}$$

$$\mathbb{E}_t\hat{y}_{t+1} = (1-z_C)\mathbb{E}_t\hat{c}_{P,t+1} + z_C\left[\bar{b}^{FI}\mathbb{E}_t\Theta_{t+1}^{\text{net}} + \bar{b}^{CB}\mathbb{E}_t\widehat{qe}_{t+1}\right]. \tag{127}$$

Subtracting (127) from (126) and applying (121) gives the IS curve:

$$\hat{y}_t = \mathbb{E}_t\hat{y}_{t+1} - \frac{1-z_C}{\sigma}\left(\hat{r}_t^s - \mathbb{E}_t\hat{\pi}_{t+1}\right) + z_C\bar{b}^{FI}\left(\Theta_t^{\text{net}} - \mathbb{E}_t\Theta_{t+1}^{\text{net}}\right) + z_C\bar{b}^{CB}\left(\widehat{qe}_t - \mathbb{E}_t\widehat{qe}_{t+1}\right). \tag{128}$$

C.4.2 Aggregate supply

Real marginal cost satisfies

$$\widehat{mc}_t = \hat{w}_t - \hat{a}_t. \tag{129}$$

Log-linearizing the labor FOC gives

$$\hat{w}_t = \chi\hat{l}_t + \sigma\hat{c}_{P,t}. \tag{130}$$

Around the zero-inflation steady state, price dispersion is second order, so

$$\hat{l}_t = \hat{y}_t - \hat{a}_t. \tag{131}$$

Goods clearing gives

$$\hat{c}_{P,t} = \frac{1}{1-z_C}\hat{y}_t - \frac{z_C}{1-z_C}\hat{c}_{B,t}. \tag{132}$$

Substituting (130), (131), and (132) into (129),

$$\widehat{mc}_t = \left(\chi + \frac{\sigma}{1-z_C}\right)\hat{y}_t - \frac{\sigma z_C}{1-z_C}\hat{c}_{B,t} - (1+\chi)\hat{a}_t. \tag{133}$$

Using (125),

$$\widehat{mc}_t = \left(\chi + \frac{\sigma}{1-z_C}\right)\hat{y}_t - \frac{\sigma z_C}{1-z_C}\left(\bar{b}^{FI}\Theta_t^{\text{net}} + \bar{b}^{CB}\widehat{qe}_t\right) - (1+\chi)\hat{a}_t. \tag{134}$$

Substituting into the linearized Calvo equation (103) yields

$$\hat{\pi}_t = \beta\mathbb{E}_t\hat{\pi}_{t+1} + \kappa_y\hat{y}_t - \Lambda_C\left(\bar{b}^{FI}\Theta_t^{\text{net}} + \bar{b}^{CB}\widehat{qe}_t\right) - \psi_a\hat{a}_t, \tag{135}$$

where

$$\kappa_y \equiv \lambda_p\left(\chi + \frac{\sigma}{1-z_C}\right), \qquad \Lambda_C \equiv \lambda_p\frac{\sigma z_C}{1-z_C}, \qquad \psi_a \equiv \lambda_p(1+\chi). \tag{136}$$

C.4.3 Credit distribution

At steady state, the adjustment term in (82) is zero and $\bar{Q}\bar{B}_i = \bar{\phi}_i \bar{v}_i$. The binding leverage constraint log-linearizes to

$$\hat{q}_t + \hat{b}_{i,t} + \eta_B(\hat{b}_{i,t} - \hat{b}_{i,t-1}) = \hat{\phi}_{i,t} + \hat{v}_{i,t}. \tag{137}$$

The bank's fundamental value can be written as

$$x_{j,t} = \Delta_t q_t B_{j,t} + \bar{x}_{j,t}, \tag{138}$$

and the transfer rule contains inherited long-bond value:

$$X^{FI}_{j,t} = \bar{X}^{FI}_j + \kappa q_t B_{j,t-1}. \tag{139}$$

The saver Euler equation prices the risk-free component, so the first-order movement in $x_{j,t}$ is

$$\hat{x}_{j,t} = s_j(\hat{\Delta}_t + \hat{q}_t + \hat{b}_{j,t}) + e_j(\hat{q}_t + \hat{b}_{j,t-1}), \tag{140}$$

where

$$s_j \equiv \frac{\bar{\Delta}\,\bar{q}\,\bar{B}_j}{\bar{x}_j}, \qquad e_j \equiv \frac{\kappa \bar{q}\,\bar{B}_j}{\bar{x}_j}. \tag{141}$$

Using $\hat{v}_{i,t} = \sum_j \mathcal{A}_{ij}\hat{x}_{j,t}$,

$$\hat{v}_{i,t} = \sum_j \mathcal{A}_{ij}\left[s_j(\hat{\Delta}_t + \hat{q}_t + \hat{b}_{j,t}) + e_j(\hat{q}_t + \hat{b}_{j,t-1})\right]. \tag{142}$$

Substituting (142) into (137) and collecting contemporaneous credit terms,

$$\begin{aligned}(1+\eta_B)\hat{b}_{i,t} - \sum_j \mathcal{A}_{ij}s_j\hat{b}_{j,t} = \hat{\phi}_{i,t} &+ \left[\sum_j \mathcal{A}_{ij}(s_j + e_j) - 1\right]\hat{q}_t \\ &+ \left[\sum_j \mathcal{A}_{ij}s_j\right]\hat{\Delta}_t + \sum_j \mathcal{A}_{ij}e_j\hat{b}_{j,t-1} + \eta_B\hat{b}_{i,t-1}.\end{aligned} \tag{143}$$

Let $\mathbf{S} = \mathrm{diag}(s_1, \ldots, s_n)$ and $\mathbf{E} = \mathrm{diag}(e_1, \ldots, e_n)$. Define

$$\boldsymbol{\alpha}_q \equiv \mathcal{A}(\mathbf{S} + \mathbf{E})\mathbf{1} - \mathbf{1}, \qquad \boldsymbol{\alpha}_\Delta \equiv \mathcal{A}\mathbf{S}\mathbf{1}. \tag{144}$$

In matrix form,

$$[(1+\eta_B)\mathbf{I} - \mathcal{A}\mathbf{S}]\,\hat{\mathbf{b}}^{FI}_t = \hat{\boldsymbol{\phi}}_t + \boldsymbol{\alpha}_q\hat{q}_t + \boldsymbol{\alpha}_\Delta\hat{\Delta}_t + (\mathcal{A}\mathbf{E} + \eta_B\mathbf{I})\hat{\mathbf{b}}^{FI}_{t-1}. \tag{145}$$

Under $\rho((1+\eta_B)^{-1}\mathcal{A}\mathbf{S}) < 1$, the inverse exists:

$$\hat{\mathbf{b}}_t^{FI} = [(1+\eta_B)\mathbf{I} - \mathcal{A}\mathbf{S}]^{-1}\left[\hat{\boldsymbol{\phi}}_t + \boldsymbol{\alpha}_q\hat{q}_t + \boldsymbol{\alpha}_\Delta\hat{\Delta}_t + (\mathcal{A}\mathbf{E} + \eta_B\mathbf{I})\hat{\mathbf{b}}_{t-1}^{FI}\right]. \tag{146}$$

The persistence matrix for inherited credit-distribution modes is

$$\mathbf{P} = [(1+\eta_B)\mathbf{I} - \mathcal{A}\mathbf{S}]^{-1}(\mathcal{A}\mathbf{E} + \eta_B\mathbf{I}). \tag{147}$$

C.4.4 Network aggregation

Define steady-state lending weights and aggregate bank credit:

$$\omega_i \equiv \frac{\bar{q}\bar{B}_i}{\sum_j \bar{q}\bar{B}_j}, \qquad \hat{b}_t^{FI} \equiv \sum_i \omega_i \hat{b}_{i,t}. \tag{148}$$

Weighting (137) by ω_i and summing yields

$$\Theta_t^{\text{net}} + \eta_B(\hat{b}_t^{FI} - \hat{b}_{t-1}^{FI}) = \hat{\phi}_t^{agg} + \hat{v}_t^{agg}, \tag{149}$$

where

$$\hat{\phi}_t^{agg} \equiv \sum_i \omega_i \hat{\phi}_{i,t}, \qquad \hat{v}_t^{agg} \equiv \sum_i \omega_i \hat{v}_{i,t}. \tag{150}$$

Expanding aggregate market value,

$$\begin{aligned} \hat{v}_t^{agg} &= \sum_i \omega_i \sum_j \mathcal{A}_{ij}\hat{x}_{j,t} \\ &= \sum_j \underbrace{\left(\sum_i \omega_i \mathcal{A}_{ij}\right)}_{\tilde{\omega}_j^\ell} \hat{x}_{j,t}. \end{aligned} \tag{151}$$

Thus

$$\begin{aligned} \hat{v}_t^{agg} &= \sum_j \tilde{\omega}_j^\ell \left[s_j(\hat{\Delta}_t + \hat{q}_t + \hat{b}_{j,t}) + e_j(\hat{q}_t + \hat{b}_{j,t-1})\right] \\ &= \left(\sum_j \tilde{\omega}_j^\ell s_j\right)\hat{\Delta}_t + \left(\sum_j \tilde{\omega}_j^\ell (s_j + e_j)\right)\hat{q}_t \\ &\quad + \sum_j \tilde{\omega}_j^\ell s_j \hat{b}_{j,t} + \sum_j \tilde{\omega}_j^\ell e_j \hat{b}_{j,t-1}. \end{aligned} \tag{152}$$

Define

$$\bar{\mu} = \sum_j \tilde{\omega}_j^\ell s_j, \qquad \bar{\alpha} = \sum_j \tilde{\omega}_j^\ell (s_j + e_j), \qquad \Lambda_j^{(0)} = \tilde{\omega}_j^\ell s_j, \qquad \Lambda_j^{(1)} = \tilde{\omega}_j^\ell e_j. \tag{153}$$

Substituting into (149) gives

$$\Theta_t^{\text{net}} = \hat{\phi}_t^{agg} + \bar{\mu}\hat{\Delta}_t + \bar{\alpha}\hat{q}_t + \mathbf{\Lambda}^{(0)}\hat{\mathbf{b}}_t^{FI} + \mathbf{\Lambda}^{(1)}\hat{\mathbf{b}}_{t-1}^{FI} - \eta_B(\hat{b}_t^{FI} - \hat{b}_{t-1}^{FI}). \tag{154}$$

C.4.5 Spread and long-bond return closure

Let $\mathcal{R} \equiv \beta/\beta_B > 1$. The saver Euler equation gives

$$\mathbb{E}_t[M_{t,t+1}^N R_t^s] = 1. \tag{155}$$

The steady spread is $\bar{\Delta} = \mathcal{R} - 1$. A first-order approximation of (73), using (155), gives

$$\hat{\Delta}_t = \frac{\mathcal{R}}{\mathcal{R} - 1}\left(\mathbb{E}_t \hat{r}_{t+1}^b - \hat{r}_t^s\right). \tag{156}$$

Log-linearizing (57) around $\bar{Q}$ gives

$$\hat{r}_{t+1}^b = \frac{\kappa\bar{Q}}{1 + \kappa\bar{Q}}\hat{q}_{t+1} - \hat{q}_t. \tag{157}$$

Under the steady-state borrower pricing condition,

$$\frac{\kappa\bar{Q}}{1 + \kappa\bar{Q}} = \beta_B\kappa,$$

so

$$\hat{r}_{t+1}^b \approx \beta_B\kappa\hat{q}_{t+1} - \hat{q}_t. \tag{158}$$